



VAJIRAM & RAVI

Institute for IAS Examination

**UPSC Preliminary
Examination
2025**

**Detailed Answer Explanation
GENERAL STUDIES - Paper I**

UPSC Prelims 2025 GS Paper - I - Answer Key			
SET - A			
1. (b)	26. (c)	51. (c)	76. (c)
2. (d)	27. (b)	52. (b)	77. (a)
3. (b)	28. (c)	53. (a)	78. (a)
4. (b)	29. (d)	54. (a)	79. (c)
5. (d)	30. (c)	55. (c)	80. (d)
6. (c)	31. (b)	56. (b)	81. (a)
7. (a)	32. (c)	57. (d)	82. (a)
8. (a)	33. (b)	58. (a)	83. (c)
9. (a)	34. (a)	59. (d)	84. (b)
10. (a)	35. (a)	60. (b)	85. (a)
11. (c)	36. (c)	61. (d)	86. (a)
12. (c)	37. (d)	62. (a)	87. (c)
13. (b)	38. (c)	63. (d)	88. (d)
14. (a)	39. (b)	64. (c)	89. (d)
15. (b)	40. (b)	65. (a)	90. (a)
16. (c)	41. (c)	66. (c)	91. (d)
17. (b)	42. (d)	67. (c)	92. (d)
18. (d)	43. (c)	68. (a)	93. (c)
19. (b)	44. (c)	69. (b)	94. (d)
20. (c)	45. (b)	70. (d)	95. (a)
21. (b)	46. (b)	71. (b)	96. (d)
22. (c)	47. (c)	72. (c)	97. (a)
23. (d)	48. (d)	73. (b)	98. (a)
24. (d)	49. (d)	74. (a)	99. (a)
25. (c)	50. (a)	75. (c)	100. (b)

Q1. With reference to investments, consider the following:

- I. Bonds
- II. Hedge Funds
- III. Stocks
- IV. Venture Capital

How many of the above are treated as Alternative Investment Funds?

- a) Only one
- b) Only two
- c) Only three
- d) All the four

Answer: b

Explanation:

- **Alternative Investment Funds (AIFs)** are privately pooled investment vehicles that collect funds from investors to invest in a variety of alternative asset classes.
- AIFs cater to investors seeking exposure to non-traditional markets and higher-risk investments. AIFs in India are regulated by the Securities and Exchange Board of India (SEBI) under the SEBI (Alternative Investment Funds) Regulations, 2012.
- These funds offer an **alternative to traditional investment options like mutual funds, bonds and stocks**, providing opportunities for higher returns and greater diversification. **So, points I and III are not correct.**
- Types of AIFs are:
 - Private Equity funds
 - **Hedge Funds. So, point II is correct.**
 - Real Estate funds
 - **Venture Capital funds. So, point IV is correct.**
 - Infrastructure funds

So, only two of the above are treated as Alternative Investment Funds. Therefore, option (b) is the correct answer.

Q2. Which of the following are the sources of income for the Reserve Bank of India?

- I. Buying and selling Government bonds
- II. Buying and selling foreign currency.
- III. Pension fund management
- IV. Lending to private companies
- V. Printing and distributing currency notes

Select the correct answer using the code given below:

- a) I and II only
- b) II, III and IV
- c) I, III, IV and V
- d) I, II and V

Answer: d

Explanation:

- The **Reserve Bank of India (RBI)** accumulates reserves out of its own income. Several of the RBI functions like giving loans to commercial banks (repo) etc. gives interest income to the RBI.
- As per statutory obligations, **this income has to be transferred to the government as the government is the single owner of the RBI.** At the same time, a certain proportion of the income

is kept apart as Contingency Fund, Asset Development Fund etc. to strengthen the RBI operations. **These funds and the valuation gains of the RBI in the form of gold, foreign currencies etc. are indicated as Capital Reserve.**

- One thing to be noted here is that **RBI is a full-service central bank**, meaning that not only does it target inflation, issue currency and regulate the banking sector, it is also the last resort lender to the government of India and the various State governments.
- The RBI can earn significant profits from some of these functions. For example, **the process of issuing currency allows for the RBI to earn something called seigniorage**. Seigniorage is basically the difference between the face value of a currency and the cost it took to produce that currency. When the RBI issues currency, say, a ₹500 note, the commercial banks have to 'buy' these notes from the central bank at the full face value (in this case, ₹500) even though it might have cost a fraction of that to actually produce that note. **So, point V is correct.**
- Then, the central bank also lends money to the Central government, State governments, and commercial banks with interest. **This interest, too, adds to the RBI's revenue.**
- Further, the investments made by the **RBI in the Government bonds** and from there on conduction of Open Market Operations by it, is also a source of revenue for the RBI. **So, point I is correct.**
- The RBI also makes investments in other countries' bonds and currencies as well, not only earning interest on these, but also **potentially benefiting from currency exchange rate fluctuations**. **So, point II is correct.**
- The Reserve Bank of India is a banker to the bankers and doesn't conduct any retail or direct consumer functions like Pension Funds Management etc. **So, point III is not correct.**
- As per the RBI Act, 1934 it can only lend to Central and State Governments, it cannot conduct any operations with the private sector, including lending to private companies. **So, point IV is not correct.**

Therefore, option (d) is the correct answer.

Q3. With reference to the Government of India, consider the information: following:

	Organization	Some of its functions	It works under
I.	Directorate of Enforcement	Enforcement of the Fugitive Economic Offenders Act, 2018	Internal Security Division-I, Ministry of Home Affairs
II.	Directorate of Revenue Intelligence	Enforces the Provisions of the Customs Act, 1962	Department of Revenue, Ministry of Finance
III.	Directorate General of Systems and Data Management	Carrying out big data analytics to assist tax officers for better policy and nabbing tax evaders	Department of Revenue, Ministry of Finance

In how many of the above rows is the information correctly matched?

- a) Only one
- b) Only two
- c) All the three
- d) None

Answer: b

Explanation:

- The **Directorate of Enforcement (ED)** is a multi-disciplinary organization mandated with investigation of offences of **money laundering** and violations of **foreign exchange laws**. It functions under the **Department of Revenue, Ministry of Finance**. ED is responsible for enforcement of the Foreign Exchange Management Act, 1999 (FEMA), Prevention of Money Laundering Act, 2002 (PMLA), and the **Fugitive Economic Offenders Act, 2018**. **So, row I is not correctly matched.**
- The **Directorate of Revenue Intelligence (DRI)** is the apex intelligence and enforcement agency for anti-smuggling efforts and primarily enforces the **Customs Act, 1962**. It works under the **Department of Revenue, Ministry of Finance**. **So, row II is correctly matched.**
- **Directorate General of Systems & Data Management**, functions under the **Central Board of Indirect taxes and Customs**, which is a statutory body under the **Department of Revenue, Government of India**. The directorate functions with the key objective to provide leadership in restructuring business processes of the department through infusion of technology, reduction of human interface with the trade & assesses and other stakeholders. It also aims at improving efficiency and transparency by implementing automation processes, besides **providing data support to the Board and field formations by compiling and disseminating data pertaining to tax collection and activities related to tax collection**, such as adjudication, appeals, court cases etc. **So, row III is correctly matched.**

So, only two of the above given rows are correctly matched.

Therefore, option (b) is the correct answer.

Q4. Consider the following statements:

I. The Reserve Bank of India mandates all the listed companies in India to submit a Business Responsibility and Sustainability Report (BRSR).

II. In India, a company submitting a BRSR makes disclosures in the report that are largely non-financial in nature.

Which of the statements given above is/are correct?

- I only
- II only
- Both I and II
- Neither I nor II

Answer: b

Explanation:

- **Business Responsibility and Sustainability Report (BRSR)**, is conceptualised in India to provide impetus to **Environmental, Social and Governance (ESG)** model in India.
- **Non-financial reporting** has been gaining traction across the world. This emphasis on nonfinancial reporting has prompted a reorientation of business models towards a more sustainable approach.
- In this backdrop, the **Securities and Exchange Board of India (SEBI)** introduced the requirement of ESG reporting in India in 2012. Their version of ESG reporting was termed the **Business Responsibility and Sustainability Report (BRSR)** and it was mandated by SEBI that the top 100 listed companies in India by market capitalization needed to file a BRSR. **So, statement I is not correct.**
- The BRSR was intended to provide a push to businesses to go above and beyond regulatory financial compliance and incorporate reporting on social and environmental impacts as well. Hence, **BRSR is a non-financial disclosure** made by business houses in India. **So, statement II is correct.**

Therefore, option (b) is the correct answer.

Q5. Consider the following statements:

Statement I:

In India, income from allied agricultural activities like poultry farming and wool rearing in rural areas is exempted from any tax.

Statement II:

In India, rural agricultural land is not considered a capital asset under the provisions of the Income-tax Act, 1961, Which one of the following is correct in respect of the above statements?

- a) Both Statement I and Statement II are correct and Statement II explains Statement I
- b) Both Statement I and Statement II are correct but Statement II does **not** explain Statement I
- c) Statement I is correct but Statement II is not correct
- d) Statement I is not correct but Statement II is correct

Answer: d

Explanation:

- **Article 366(1)** of the Constitution of India provides that the expression 'agricultural income' in the Constitution means **agricultural income as defined for the purpose of enactments relating to Indian Income Tax.**
- **Taxes on agricultural income falls under Entry 46 in "State List" under the Constitution of India.** Thus, only the State Governments are competent to enact legislations for taxation of agricultural income. The Central Government cannot levy income tax on agricultural income. However, agricultural income is considered for rate purposes while determining the income tax liability.
- As per section 2(1A) of the Income Tax Act, 1961 '**agricultural income**' means
 - o Any rent or revenue derived from land which is situated in India and is used for agricultural purposes
 - o Any income derived from such land by agricultural operations including processing of agricultural produce so as to render it fit for market or sale of such produce;
 - o Any income attributable to a farm house subject to fulfilment of conditions specified in the Act; and
 - o Any income derived from saplings or seedlings grown in a nursery
- To sum up, for the purpose of Income Tax act, 1961 **agricultural income is any rent or revenue by means of cash or in kind, derived from a land, which is used for an agricultural purpose and land should be situated in India.**
- The Supreme Court in the **Commissioner of Income Tax vs. Raja Benoy Kumar Suhas Roy, 1957** provided a definition for 'agricultural processes'. In this judgement the court said that the mere fact that an activity has some connection with or is in some way dependent on land is not sufficient to bring it within the scope of the term and such extension of the term "agriculture" is unwarranted. Therefore, **activities like dairy farming, poultry farming, cheese & butter making or wool rearing cannot be taken to as agricultural purposes for the purpose of calculating the agricultural income. so, statement I is not correct.**
- It should be noted here that the Central Government has recently presented a bill in the parliament, which amends the Income Tax act. In this bill it is explicitly stated that income from dairy farming, poultry, and fisheries will not be considered an 'agricultural income' (up until now, the IT act has largely been

silent on it and for the questions regarding what constitutes agricultural purposes are being governed by the principles laid forth in **Commissioner of Income Tax vs. Raja Benoy Kumar Suhas Roy, 1957**).

- The term 'capital asset' has been defined in **Section 2(14) of the Income-tax Act** as, property of any kind held by an assessee, whether or not connected with his business or profession or any securities held by an individual or institution in accordance with the Security and Exchange Board of India's Regulations.
- It explicitly exempts Agricultural land situated in any rural area in India. so, statement II is correct.

So, statement I is not correct but Statement II is correct

Therefore, option (d) is the correct answer.

Q6. Consider the following statements:

I. India has joined the Minerals Security Partnership as a member.

II. India is a resource-rich country in all the 30 critical minerals that it has identified.

III. The Parliament in 2023 has amended the Mines and Minerals (Development and Regulation) Act, 1957 empowering the Central Government to exclusively auction mining lease and composite license for certain critical minerals.

Which of the statements given above are correct?

- I and II only
- II and III only
- I and III only
- I, II and III

Answer: c

Explanation:

- **Critical minerals** are defined as a set of naturally occurring elements and compounds that have diverse irreplaceable industrial applications but confront supply-related vulnerabilities either in the form of their limited geographic occurrences or sourcing challenges.
- The **Mineral Security Partnership (MSP)** is a US-led plurilateral partnership, which, according to its guiding principles, "strives to promote responsible growth across the critical minerals sector via a shared commitment to high Environmental, Social, and Governance (ESG) standards; sustainability; and shared prosperity."
- The **MSP currently consists of 23 partners**, including Argentina, Australia, Canada, Estonia, Finland, France, Germany, Greenland, **India**, Italy, Japan, Kazakhstan, Mexico, Namibia, Norway, Peru, the Republic of Korea, Sweden, Ukraine, the UK, the US, Uzbekistan, and the EU (represented by the European Commission). **So, statement I is correct.**
- Based on inter-ministerial consultations and adopted criteria which included the economic importance and supply chain risks associated with minerals, **the Ministry of Mines identified and released a list 30 critical minerals in June 2023**. Out of these thirty elements, fourteen have been highlighted for having both high economic importance as well as high supply risks. Seven have been categorized as elements of high supply risks and nine as elements with high economic importance.
- As stated, supply chain vulnerabilities, formed one of the criteria for the listing of minerals. **India did have significant reserves for some of the listed critical minerals like Copper, Graphite etc. but is largely import dependent for others like Antimony, Germanium etc. So, statement II is not correct.**

- As per the provisions of the **Mines and Minerals (Development and Regulation) Act, 1957 (MMDR Act)**, mineral concessions are auctioned by the State Governments. Some of the minerals identified as critical minerals were in the list of atomic minerals under the MMDR Act and hence their exploration and mining was reserved only for Government companies. In order to prioritise their exploration and mining, the **MMDR Act was amended in August 2023**. Through the amendment, 6 minerals, viz., lithium, beryllium, titanium, niobium, tantalum and zirconium, were removed from the list of atomic minerals thereby enabling the participation of private entities in exploration and mining of these critical minerals.
- Further, a new list of 24 critical and strategic minerals (Appendix) was added in the MMDR Act which also includes the 6 minerals removed from the list of atomic minerals. **The Central Government has been empowered to auction the blocks for these critical and strategic minerals**. The Central Government can now provide for Exploration and **Composite Licenses (CL)** for these critical minerals exclusively. **So, statement III is correct.**

Therefore, option (c) is the correct answer.

Q7. Consider the following statements:

Statement I:

As regards returns from an investment in a company, generally, bondholders are considered to be relatively at lower risk than stockholders.

Statement II:

Bondholders are lenders to a company whereas stockholders are its owners.

Statement III :

For repayment purpose, bond-holders are prioritized over stockholders by a company. Which one of the following is correct in respect of the above statements?

- Both Statement II and Statement III are correct and both of them explain Statement I
- Both Statement I and Statement II are correct and Statement I explains Statement III
- Only one of the Statements II and III is correct and that explains Statement I
- Neither Statement II nor Statement III is correct

Answer: a

Explanation:

- **Stocks represent ownership shares in public corporations** and are a fundamental component of the equity market. **When an investor purchases a stock, they acquire a portion of ownership in the company**, entitling them to a share of its assets and earnings, along with the liabilities.
- **Bonds are debt securities that can be traded**. Bonds, in essence are a promise from the issuer party to pay compensation in the form of interest in a certain period and pay off the principal at the end of a predetermined time, to the buyer of the bonds.
- Bonds are a fixed income stock investment that aims to provide a relatively stable growth rate of investment value with relatively more stable risk compared to stocks. **So, statement I is correct.**
- **Owning stocks of an entity makes the investor part owner of that entity**, and the money paid up by the investor becomes part of the capital for the entity, **thus making the investor a claimant to the profits as well as sharer of the liabilities of the entity**.
- While by owning the **bonds, the investor is no more than a lender to the entity**, money raised via bonds are not classified as fixed capital but as liability to the entity, which it is obligated to return as per the terms and conditions of the bond. **So, statement II is correct.**

- On the virtue of being the part owners of the entity, stockholders become liable towards the obligations which the entity pre-holds or takes up in future, which also includes borrowings via bonds.
- As paying back bonds is a legal obligation, thus **in case of insolvency bond owners are prioritized over stockholders for payment. So, statement III is correct.**

So, both Statement II and Statement III are correct and both of them explain Statement I. Therefore, option (a) is the correct answer.

Q8. Consider the following statements:

I. India accounts for a very large portion of all equity option contracts traded globally thus exhibiting a great boom.

II. India's stock market has grown rapidly in the recent past even overtaking Hong Kong's at some point of time.

III. There is no regulatory body either to warn the small investors about the risks of options trading or to act on unregistered financial advisors in this regard.

Which of the statements given above are correct?

- I and II only
- II and III only
- I and III only
- I, II and III

Answer: a

Explanation:

- As per recent statistics, **India is the fifth-largest country globally in terms of market capitalization**, trailing behind the United States (\$55 trillion), China (\$9 trillion), Japan (\$6.1 trillion), and Hong Kong (\$5.5 trillion). This increased market share of India in equity markets has been the cause and effect of Indian markets contributing to a greater portion of equity option contracts. **So, statement I is correct.**
- **India surpassed Hong Kong's market value to become the fourth largest economy by market capitalization for the first time in January, 2024** and again in June 2024 before losing out in July of the same year. **So, statement II is correct.**
- **Economic Survey 2024-25**, raised concerns over large numbers of retail and small investors becoming participants in the market, thus creating a situation of asset bubble, and their savings becoming subjected to large scale market volatilities.
- In India, the mandate of **Securities and Exchange Board of India (SEBI)**, a statutory body under the SEBI act, 1992 is to **protect the interests of investors in securities** and to regulate the securities market and for matters connected therewith or incidental thereto, this includes warning the small investors about the risks of options trading or to act on unregistered financial advisors. **So, statement III is not correct.**

Therefore, option (a) is the correct answer.

Q9. Consider the following statements:

Statement I:

Circular economy reduces the emissions of greenhouse gases.

Statement II:

Circular economy reduces the use of raw materials as inputs.

Statement III:

Circular economy reduces wastage in the production process.

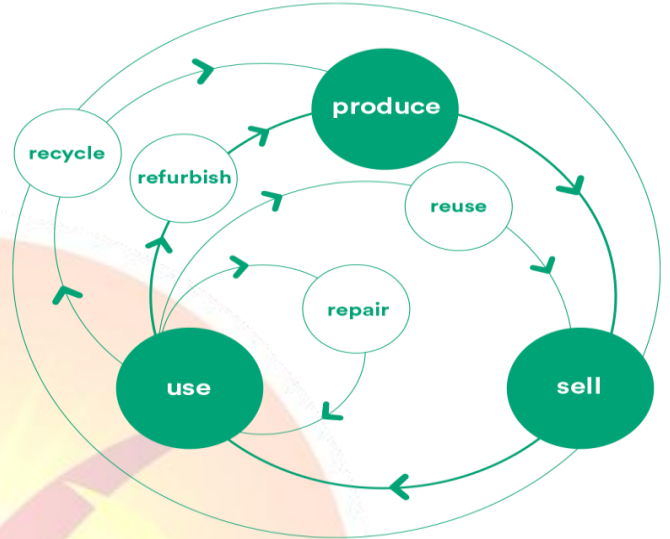
Which one of the following is correct in respect of the above statements?

- a) Both Statement II and Statement III are correct and both of them explain Statement I
 b) Both Statement II and Statement III are correct but only one of them explains Statement I
 c) Only one of the Statements II and III is correct and that explains Statement I
 d) Neither Statement II nor Statement III is correct

Answer: a

Explanation:

- **Circular economy** refers to a system that aims to minimize the use of energy, natural resources and waste generation by closing loops of utilities and materials flows. **It involves minimizing inputs of raw materials, reducing waste outputs and reintegrating products into the system when they reach the end of their life.**
- It works by extending product lifespan through improved design and servicing and relocating waste from the end of the supply chain to the beginning—in effect, **using resources more efficiently by using them over and over, not only once.**
- In practice, it implies reducing waste to a minimum. When a product reaches the end of its life, its materials are kept within the economy wherever possible thanks to recycling. These can be productively used again and again, thereby creating further value.



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- **This is a departure from the traditional, linear economic model**, which is based on a take-make-consume-throw away pattern. This model relies on large quantities of cheap, easily accessible materials and energy.
- A major benefit and a cause for call towards the circular economy is a **reduction in total annual Greenhouse Gas (GHGs) emissions**. **So, statement I is correct.**
- Creating more efficient and sustainable products from the start would **help to reduce energy and resource consumption, thus lesser greenhouse gases are emitted as a part of production process**. **So, statement II is correct.**
- As the **concept of circular economy revolves around the wastes of one product cycle acting as raw materials for another**, thereby it leads to reduction in waste generation as well, as waste in today's day and age is a major contributor of GHG emissions, like methane, adoption of circular economy will further lead to reduced warming because of GHGs. **So, statement III is correct.**

So, both Statement II and Statement III are correct and both of them explain Statement I. Therefore, option (a) is the correct answer.

Q10. Consider the following statements:

I. Capital receipts create a liability or cause a reduction in the assets of the Government.

II. Borrowings and disinvestment are capital receipts.

III. Interest received on loans creates a liability of the Government.

Which of the statements given above are correct?

- I and II only
- II and III only
- I and III only
- I, II and III

Answer: a

Explanation:

- In the **balance of payment** accounts, receipts are classified as capital or revenue. **Capital receipts** refer to incoming cash flows (receipts) originating from the sale of fixed assets, shares, or debt and are generally non-recurring.
- **Capital receipts leads to alteration in assets or liabilities of the receiver**, on the other hand revenue receipts being day to day in nature are not asset or liability creating. **So, statement I is correct.**
- **Borrowings and disinvestment are classic examples of capital receipts**. Borrowings lead to future repayment obligations, while disinvestment involves selling public assets, thus reducing government holdings. **So, statement II is correct.**
- **Interest received on loans is revenue income for the government and does not result in any liability**. It is a return on the capital lent by the government and is thus a revenue receipt, not a capital one. **So, Statement III is not correct.**

Therefore, option (a) is the correct answer.

Q11. Consider the following statements about Raja Ram Mohan Roy :

I. He possessed great love and respect for the traditional philosophical systems of the East.

II. He desired his countrymen to accept the rational and scientific approach and the principle of human dignity and social equality of all men and women.

Which of the statements given above is/are correct?

- a) I only
- b) II only
- c) Both I and II
- d) Neither I nor II

Answer: c

Explanation:

- **Raja Ram Mohan Roy** was a central figure in the Indian Socio-Religious Reform Movement of the early 19th century and is regarded as the first great leader of Modern India.
- He studied Persian-Arabic literature, and later came under the influence of Vedic monism. **He tried to improve Hinduism in the light of modern scientific temper and logical thinking and issued a stinging criticism of idolatry, polytheism and clericalism.**
- Despite being critical of certain aspects of Hinduism Raja Ram Mohun Roy had **great respects for traditional philosophical systems**, to the extent that he advocated for reforms in Hinduism by going back to its purist form as enshrined in Vedanta text. **So, statement I is correct.**
- Despite being an admirer of Western rationalism, scientific temper and enlightenment, he was equally critical of Western religion as well. He **only aspired for inculcating the rationalist, scientific approach of the west to bring the Indian society out of stagnation**, so that human and dignity and social equality can be ensured for all men and women. **So, statement II is correct.**

Therefore, option (c) is the correct answer.

Q12. Consider the following subjects with regard to Non-Cooperation Programme:

- I. Boycott of law-courts and foreign cloth
- II. Observance of strict non-violence
- III. Retention of titles and honours without using them in public
- IV. Establishment of Panchayats for settling disputes

How many of the above were parts of Non-Cooperation Programme?

- a) Only one
- b) Only two
- c) Only three
- d) All the four

Answer: c

Explanation:

- The **Non-Cooperation Movement (NCM)**, was a Gandhian **mass movement** beginning in January 1921 in the background of Khilafat movement and the Jallianwala Bagh incident. The philosophy behind the movement was if the Indians collectively refuse to cooperate with the British, it will become untenable for them to continue governing the country and have to eventually cede to the demands of Indians.
- The movement, Gandhi assured, would bring Swaraj within one year. If that did not happen or if Government resorted to oppression a Civil Disobedience campaign would be launched.
- In the **NCM, two types of programs were arranged the negative and constructive programs.** In **negative program** following were included:
 - **Renunciation of the titles and state honors. So, point III is not correct.**

- Boycott of schools and colleges associated with the government.
- **Boycott of Courts and foreign goods. So, point I is correct**
- On the other hand, constructive programs were also adopted such as:
 - Establishment of national educational institutions,
 - **Establishment of village panchayats. So, point IV is correct.**
 - Promotion of spinning-weaving.
- All this were to be followed while laying **strict emphasis and following of non-violence. So, point II is correct.**
- In the later stage of the movement these measures were to extended to the stage of Civil Disobedience, which would also involve resignation from government service and No tax movement.

So, only three of the above were parts of Non-Cooperation Programme.

Therefore, option (c) is the correct answer.

Q13. The irrigation device called 'Araghatta' was

- a) a water bag made of leather pulled over a pulley
- b) a large wheel with earthen pots tied to the outer ends of its spokes
- c) a larger earthen pot driven by bullocks
- d) a large water bucket pulled up by rope directly by hand

Answer: b

Explanation:

- The word '**araghatta**' comes from the combination of two words '**ara**,' meaning spokes, and '**ghatta**' meaning pot. Therefore, araghatta becomes a **system of spokes with a pot at the end.**
- **Araghatta** was an irrigation device, with a **rope-pot system of lifting water from open wells.** In the modern English literature, it is referred to as **Persian wheel**, probably because of its first discovery by European historians in Persia, but the use of *araghatta* in India dates past the use in Persia.

Therefore, option (b) is the correct answer.

Q14. Who among the following rulers in ancient India had assumed the titles 'Mattavilasa', 'Vichitrachitta' and 'Gunabhara'?

- a) Mahendravarman I
- b) Simha Vishnu
- c) Narasimhavarman I
- d) Simhavarman

Answer: a

Explanation:

- **Mahendravarman I** was the Pallava king who reigned from c. 600 to 630 AD, he was a great patron of arts and literatures and was responsible for **building temples at Panamalai, Mandagapattu and Kanchipuram.**
- He was a follower of Jainism in the early part of his career and had turned to Shaivism under the influence of Shaiva saint Thirunavukkarasar alias Appar. It was after this conversion that the king wrote his famous one act play '**Mattavilasa Paramhansa**', revolving around an ascetic Kapilka who was unjustly accused by the two protagonists for having stolen a skull from them. From here on **the title of Mattvilasa got associated with Mahendravarman I.**

- The inscription at Mandagapattu mentions **Mahendravarman I** with numerous titles such as **Vichitrachitta (curious-minded)**, **Chitrakarapuli (tiger among artists)**, **Chaityakari (temple builder)**, which show his interest in art activities.
- Later in his reign he also assumed the title of **Gunabhara** (full of virtues) for himself.

Therefore, option (a) is the correct answer.

Q15. Fa-hien (Faxian), the Chinese pilgrim travelled to India during the reign of

- Samudra Gupta
- Chandragupta II
- Kumaragupta I
- Skandagupta

Answer: b

Explanation:

- **Fa-Hein (Faxian)** was one of the earliest Chinese Buddhist pilgrims to visit India, visiting the country during the **4th century AD**. He was over 60 years old when he left Chang'an on his long overland journey to India and about 77 when he returned to China. His main aim was to obtain and bring back texts containing monastic rules.
- He left his travelogues in the book **Gaoseng Faxian zhuan (A Record of Buddhist Kingdoms)**, the earliest first-hand Chinese account of Buddhist sites and practices in India.
- At the time of Fa Xian's visit to the country **the Gupta king Chandragupta II** was reigning in Pataliputra. Although the book doesn't contain the name of the reigning king, it contains several observations about the life of the people of the subcontinent.

Therefore, option (b) is the correct answer.

Q16. Who among the following led a successful military campaign against the kingdom of Srivijaya, the powerful maritime State, which ruled the Malay Peninsula, Sumatra, Java and the neighbouring islands?

- Amoghavarsha (Rashtrakuta)
- Prataparudra (Kakatiya)
- Rajendra I (Chola)
- Vishnuvardhana (Hoysala)

Answer: c

Explanation:

- **Rajendra I** was a ruler of the **imperial Chola line**, assuming the throne from his father Rajaraja I in **c. 1014AD**, he continued the process of territorial expansion of the Cholas. The Chola empire reached its territorial zenith during his reign.
- While the earlier Chola kings expanded the territories over the land, Rajaraja adopted an aggressive naval campaign and brought many offshore territory under his suzerainty.
- In 1025 AD he dispatched a successful naval campaign to the **kingdom of Sri Vijaya in the Malaya peninsula** subsuming the territories of today's **Sumatra, Java and other neighbouring islands**.
- Owing to his aggressive naval expansion, the Bay of Bengal was started to be called the Chola lake during his reign.

Therefore, option (c) is the correct answer.

Q17. With reference to ancient India (600-322 BC), consider the following pairs:

	Territorial region	River flowing in the region
I.	Asmaka	: Godavari
II.	Kamboja	: Vipasa
III.	Avanti	: Mahanadi
IV.	Kosala	: Sarayu

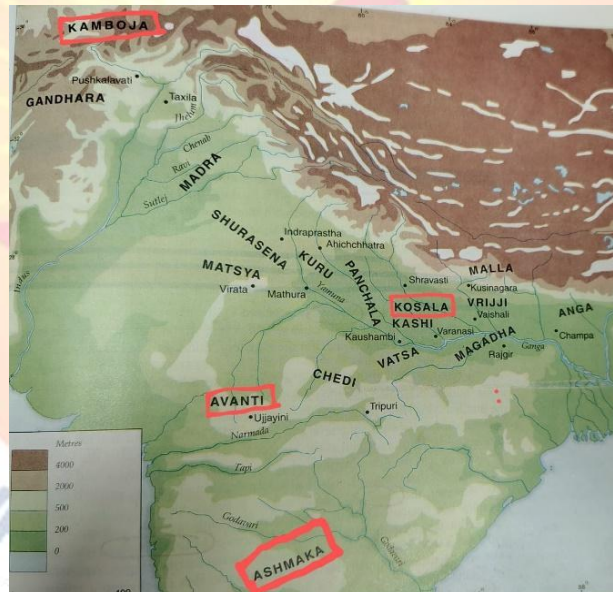
How many of the pairs given above are correctly matched?

- Only one
- Only two
- Only three
- All the four

Answer: b

Explanation:

- During the 6th-5th century BC state polities and societies emerged in the belt stretching from Gandhara in the North-West to Godavari in the South. The Puranic along with the Buddhist and Jaina texts mention the emergence of **16 Mahajanapadas** (great states) in the subcontinent.
- These *Mahajanapadas* were centred around the deltas of major Rivers, signifying the importance of agricultural activities in the states. Also, these Rivers acted as natural frontiers of defence for these states.



- Asmaka:** This *Mahajanapada* was located on the banks of the **river Godavari**. Its capital was Potana/Potali, identified with modern Botan. **So, pair I is correctly matched**
- Kamboja:** It included the area around Rajouri, including the Hazara district of the North-West Frontier province of Pakistan. **So, pair II is not correctly matched**
- Avanti:** It was located in the Malwa region, and was divided into two regions – Mahishmati and Ujjayini, both of which are mentioned as its capital. The **Shipra river** flowed along the kingdom. **So, pair III is not correctly matched**
- Kosala:** The kingdom of Kosala was bounded by the Gomati on the west, the Sarpika or Syandika (Sai) on the south, and the Nepal hills to the north. **The Sarayu river divided it into a northern and a southern part** – Shravasti (identified with modern Saheth-Maheth) was the capital of north Kosala, and

Kushavati the capital of south Kosala. Saketa and Ayodhya were two other important towns and may once have been political centres. **So, pair IV is correctly matched.**

So, only two of the pairs given above are correctly matched.

Therefore, option (b) is the correct answer.

Q18. The first Gandharva Mahavidyalaya, a music training school, was set up in 1901 by Vishnu Digambar Paluskar in:

- a) Delhi
- b) Gwalior
- c) Ujjain
- d) Lahore

Answer: d

Explanation:

- **Vishnu Digambar Paluskar** was born on August 18, 1872 to a Maharashtrian family in the town of Kurundwad near Pune. He was a talented and trained Hindustani musician, and was of the vision to **make teaching of music egalitarian**, which was until then restricted to Gharanas.
- This made him conceptualize the idea of Gandharva Mahavidyalayas, with an **objective to teach music to any aspirant without the necessity of being associated with a guru and gharana**.
- In those days, **Lahore** was the seat of culture, making it the ideal location for Paluskar to open the **Akhil Bharatiya Gandharva Mahavidyalaya**, the first institution imparting training in music that was supported by public contributions rather than royal patronage.
- **Paluskar**, is also known for his composition of the song Raghupati Raghav Raja Ram, a rendition of which was made popular by Mahatma Gandhi.



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Therefore, option (d) is the correct answer.

Q19. Ashokan inscriptions suggest that the 'Pradeshika', 'Rajuka' and Yukta' were important officers at the

- a) village-level administration
- b) district-level administration
- c) provincial administration
- d) level of the central administration

Answer: b

Explanation:

- The Mauryans maintained an illustrious and elephantine bureaucracy, right from the central to district level. This bureaucracy was further illustrated during the Ashokan times, when the empire reached its territorial zenith .
- **Ashokan inscriptions suggest that the *pradeshika*, *rajuka*, and *yukta* were important officers at the district level.**
- **Rock edict 3 refers to these officers going on tours every five years in order to instruct people in dhamma and for other purposes.**
- The term *rajuka* may come from *rajju*, meaning rope, and the reference may be to the measurement of land using ropes. **Though land measurement may have been their main or original duty, in Ashoka's time, the rajukas seem to have been high-ranking officers who were also associated with public welfare measures.** Ashoka added judicial functions and *dhamma* propagation to their duties.
- The term *yukta* occurs in the Arthashastra as a general term for officers, during the Ashokan times yuktas assisted the sthanikas and gopas in revenue collection.

Therefore, option (b) is the correct answer.

Q20. Consider the following statements in respect of the Non-Cooperation Movement:

- I. The Congress declared the attainment of 'Swaraj' by all legitimate and peaceful means to be its objective.
- II. It was to be implemented in stages with civil disobedience and non-payment of taxes for the next stage only if 'Swaraj' did not come within a year and the Government resorted to repression.

Which of the statements given above is/are correct?

- a) I only
- b) II only
- c) Both I and II
- d) Neither I nor II

Answer: c

Explanation:

- The **Non-Cooperation Movement (NCM)**, was a Gandhian **mass movement** beginning in January 1921 in the background of Khilafat movement and the Jallianwala Bagh incident. The philosophy behind the movement was if the Indians collectively refuse to cooperate with the British, it will become untenable for them to continue governing the country and have to eventually cede to the demands of Indians.
- In opposition to the promulgation of Rowlatt act Gandhi had earlier launched the Rowlatt Satyagraha as a nationwide mass movement with the principles of non-cooperation and non-violence, but Gandhi had

to withdraw it because of weak grassroot leadership. It is where Gandhi realised the need of an All India Political organization like the Congress.

- **Congress at that time was vary of the idea of such a mass movement**, established leaders like CR Dass believed that such a move will delay the implementation of constitutional progress.
- The proposal of Gandhi did got approved from the Congress at a special session Convened at Calcutta in September 1920, where the NCM was approved.
- Later in December that year at its full session at Nagpur, Congress again endorsed the proposal and **committed for the attainment of 'Swaraj' by all legitimate and peaceful means to be its objective. So, statement I is correct.**
- The movement, **Gandhi assured, would bring Swaraj within one year. If that did not happen or if Government resorted to oppression a Civil Disobedience campaign would be launched. So, statement II is correct.**

Therefore, option (c) is the correct answer.

Q21. Consider the following countries:

- I. Austria
- II. Bulgaria
- III. Croatia
- IV. Serbia
- V. Sweden
- VI. North Macedonia

How many of the above are members of the North Atlantic Treaty Organization?

- a) Only three
- b) Only four
- c) Only five
- d) All the six

Answer: b

Explanation:

- **North Atlantic Treaty Organization (NATO)**, is a cold war era military alliance **established in 1949** between the Western European and North American countries (USA and Canada) that sought to create a counterweight to Soviet armies stationed in central and eastern Europe after World War II.
- It is a collective security agreement, wherein any armed attack on any of the participating country is taken as attack on all of the NATO members.
- **Article 10 of the North Atlantic Treaty sets out how countries can join the Alliance.** It states that membership is open to any "European State in a position to further the principles of this Treaty and to contribute to the security of the North Atlantic area".
- **Any decision to invite a country to join the Alliance is taken by the North Atlantic Council, NATO's principal political decision-making body, on the basis of consensus among all Allies.**
- **NATO currently has 32 members across Europe and North America** - the original 12 founders plus 20 countries which have joined since 1949. After the Soviet Union's collapse in 1991, many Eastern European countries became members, including Albania, Bulgaria, the Czech Republic, Estonia, Hungary, Latvia, Lithuania, Poland, Romania and Slovakia.
- Current members of the NATO are:
 - o Albania
 - o Belgium

- o **Bulgaria. So, point II is correct**
- o Canada
- o **Croatia. So, point III is correct.**
- o Czech Republic
- o Denmark
- o Estonia
- o Finland
- o France
- o Germany
- o Greece
- o Hungary
- o Iceland
- o Italy
- o Latvia
- o Lithuania
- o Luxembourg
- o Montenegro
- o Netherlands
- o **North Macedonia. So, point VI is correct.**
- o Norway
- o Poland
- o Portugal
- o Romania
- o Slovakia
- o Slovenia
- o Spain
- o **Sweden. So, point V is correct.**
- o Turkey
- o United Kingdom
- o United States.

- **Sweden and Austria are not members of the NATO. So, point I and IV are not correct.**

So, only four of the above are members of the North Atlantic Treaty Organization. Therefore, option (b) is the correct answer.

Q22. Consider the following countries:

- I. Bolivia
- II. Brazil
- III. Colombia
- IV. Ecuador
- V. Paraguay
- VI. Venezuela

Andes mountains pass through how many of the above countries?

- a) Only two
- b) Only three

c) Only four

d) Only five

Answer: c

Explanation:

- The Andes is **the longest mountain range in the world** and are second only to Himalayas in elevation.
- Spreading across the continent of South America their expanse is spread across seven countries:
 - Argentina
 - **Bolivia. So, point I is correct.**
 - Chile
 - **Colombia. So, point III is correct.**
 - **Ecuador. So, point IV is correct.**
 - Peru
 - **Venezuela. So, point VI is correct.**
- The Andes doesn't extend to the countries of Brazil and Paraguay which are not located on the western margin of the continent. **So, points II and V are not correct.**

So, the Andes mountains pass through only four of the above countries. Therefore, option (c) is the correct answer.



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Q23. Consider the following water bodies:

- I. Lake Tanganyika
- II. Lake Tonle Sap
- III. Patos Lagoon

Through how many of them does the equator pass?

- a) Only one
- b) Only two
- c) All the three
- d) None

Answer: d

Explanation:

- **Lake Tanganyika:** It is one of the three and the largest lake amongst the **African Great Rift Valley lakes**. It is also the second deepest and the second largest lake in the world by volume. It extends across the countries of Tanzania, Democratic Republic of Congo, Burundi and Zambia. **It lies along the 6.25° S latitude, south of the Equator. So, point I is not correct.**
- **Lake Tonle Sap:** It is the largest and most important natural freshwater lake in Southeast Asia, and flows entirely in the country of **Cambodia**. It plays a critical role in Cambodian Khmer culture, in the economy of Cambodia, and in the ecology and hydrology of the Mekong River basin. **It is located at the latitude of 13°N, lying north to the equator. So, point II is not correct.**
- **Patos Lagoon:** It is the largest choked lagoon in the world, extending from the city of Porto Alegre (30°30' S) to Rio Grande (32°12' S) in southern **Brazil**, at the Atlantic coast of South America. It lies South of the Equator. **So, point III is not correct.**

So, none of the above given points are correct.

Therefore, option (d) is the correct answer.

Q24. Consider the following statements about turmeric during the year 2022-23:

- I. India is the largest producer and exporter of turmeric in the world.
- II. More than 30 varieties of turmeric are grown in India.
- III. Maharashtra, Telangana, Karnataka and Tamil Nadu are major turmeric producing States in India.

Which of the statements given above are correct?

- a) I and II only
- b) II and III only
- c) I and III only
- d) I, II and III

Answer: d

Explanation:

- **Turmeric** is a product of *Curcuma longa*, a **rhizomatous herbaceous perennial plant** belonging to the ginger family Zingiberaceae, which is **native to tropical South Asia**.
- **India** produces nearly 62% all of the world's turmeric crop and consumes 80% of it, making it the **largest producer, consumer and exporter of Turmeric in the world**. With its inherent qualities and high content of the important bioactive compound curcumin, Indian turmeric is considered to be the best in the world. **So, statement I is correct.**
- **More than 30 varieties of Turmeric are grown in India** and it is grown in over 20 states in the country. **So, statement II is correct.**

- The largest producing states of Turmeric in India are Maharashtra, Telangana, Karnataka and Tamil Nadu. So, statement III is correct.

Therefore, option (d) is the correct answer.

Q25. Which of the following are the evidences of the phenomenon of continental drift?

- The belt of ancient rocks from Brazil coast matches with those from Western Africa.
- The gold deposits of Ghana are derived from the Brazil plateau when the two continents lay side by side.
- The Gondwana system of sediments from India is known to have its counterparts in six different landmasses of the Southern Hemisphere.

Select the correct answer using the code given below:

- I and III only
- I and II only
- I, II and III
- II and III only

Answer: c

Explanation:

- The **Continental Drift Theory**, proposed by **Alfred Wegener** in 1912, revolutionized our understanding of Earth's geological history. It posits that continents were once united in a supercontinent called Pangaea and have since drifted apart over geological time.
- He gave evidences in support of his theory as:
 - Similar rock types and structures are found on continents now separated by vast oceans. For instance, the Appalachian Mountains in North America align geologically with the Caledonian Mountains in Scotland and Scandinavia.
 - The **matching ancient rock belts between Brazil and Western Africa are a well-documented geological evidence of continental drift**. These rocks share identical age, composition and structure, supporting the idea that South America and Africa were once joined. **So, statement I is correct.**
 - Geological studies suggest that the **gold deposits in Ghana might have originated from the Brazilian Plateau**, which aligns with the theory that the continents were once connected. **So, statement II is correct.**
 - The **Gondwana sedimentary system found in India is also found in other southern landmasses like Africa, South America, Antarctica and Australia**, providing strong evidence of past continental connection. **So, statement III is correct.**

Therefore, option (c) is the correct answer.

Q26. Consider the following statements:

Statement I:

The amount of dust particles in the atmosphere is more in subtropical and temperate areas than in equatorial and polar regions.

Statement II:

Subtropical and temperate areas have less dry winds.

Which one of the following is correct in respect of the above statements?

- Both Statement I and Statement II are correct and Statement II explains Statement I
- Both Statement I and Statement II are correct but Statement II does not explain Statement I
- Statement I is correct but Statement II is not correct
- Statement I is not correct but Statement II is correct

Answer: c

Explanation:

- Atmosphere has a sufficient capacity to keep small solid particles, which may originate from different sources and include sea salts, fine soil, smoke-soot, ash, pollen, dust and disintegrated particles of meteors.
- **Dust particles are generally concentrated in the lower layers of the atmosphere;** yet, convectional air currents may transport them to great heights.
- **The higher concentration of dust particles is found in subtropical and temperate regions** as compared to that in the equatorial regions. **So, statement I is correct.**
- The amount of dust particles directly varies with that of humidity in the air; the lower the humidity, the higher is the concentration of suspended dust particles.
- As the equatorial and polar regions are characterised by humid air, there are less amount of suspended dust particles in the atmosphere, while **the temperate and subtropical regions, owing to drier winds in comparison to equatorial and polar regions,** have higher concentrations of suspended dust particles. **So, statement II is not correct.**

So, statement I is correct but Statement II is not correct.

Therefore, option (c) is the correct answer.

Q27. Consider the following statements:

Statement I:

In January, in the Northern Hemisphere, the isotherms bend equatorward while crossing the landmasses, and poleward while crossing the oceans.

Statement II:

In January, the air over the oceans is warmer than that over the landmasses in the Northern Hemisphere.

Which one of the following is correct in respect of the above statements?

- Both Statement I and Statement II are correct and Statement II explains Statement I
- Both Statement I and Statement II are correct but Statement II does **not** explain Statement I
- Statement I is correct but Statement II is not correct
- Statement I is not correct but Statement II is correct

Answer: b

Explanation:

- The **isotherms** are lines joining places having equal temperature. In general, the effect of the latitude on temperature is well pronounced on the map, as the isotherms are generally parallel to the latitude.
- In the **northern hemisphere, the land surface area is much larger than in the southern hemisphere.** Hence, the effects of land mass and the ocean currents are well pronounced. **In January, the isotherms deviate to the north, that is poleward, over the ocean and to the south, that is towards the equator, over the continent.**
- During the winter months in the Northern hemisphere, observe marked temperature variations over the land and the oceans. **Oceans have a high heat capacity compared to land,** meaning they can retain more heat energy with minimal temperature changes. This makes the ocean a thermal buffer, moderating temperature fluctuations over both short and long periods.
- **Air temperatures over land fluctuate more because land changes its temperature much more rapidly than ocean water does.** That is why the isotherms over land bend fluctuate more compared to the ones over the Oceans. **So, statement II is correct.**

So, both Statement I and Statement II are correct but Statement II does not explain Statement I.

Therefore, option (b) is the correct answer.

Q28. Consider the following statements:

Statement I:

In the context of effect of water on rocks, chalk is known as a very permeable rock whereas clay is known as quite an impermeable or least permeable rock.

Statement II:

Chalk is porous and hence can absorb water.

Statement III :

Clay is not at all porous.

Which one of the following is correct in respect of the above statements?

- Both Statement II and Statement III are correct and both of them explain Statement I
- Both Statement II and Statement III are correct but only one of them explains Statement I
- Only one of the Statements II and III is correct and that explains Statement I
- Neither Statement II nor Statement III is correct

Answer: c

Explanation:

- Chalk is a soft, white, porous, sedimentary rock composed primarily of the mineral calcite (calcium carbonate).** It is often associated with marine environments and is characterized by its distinctive white color and powdery texture.
- It is primarily composed of the remains of microscopic marine organisms known as coccolithophores. These organisms secrete calcium carbonate shells or plates, and when they die, their remains accumulate on the ocean floor. Over time, these accumulated remains can form thick layers of chalk rock.
- Porosity and permeability of the soil depend on the particle size of the soil** – if the sediments or rock particles are composed of very small grains, such as in clays and silts, the space through which water can flow is limited. In addition, clay particles have a lot of surface area to which hygroscopic water attaches, creating a further resistance to fluid movement.
- If sediments are comprised of coarser grains, like chalk stone, the pore space is more available. **These coarser grains also have less surface area, so less water can attach to them, allowing better fluid movement. So, statement I is correct.**
- Being a porous rock, **chalk can retain higher moisture as water molecules can seep into these pores**, in addition to this the calcite salts are known to have good adsorption properties. **So, statement II is correct.**
- Although clay is considered less porous because of fine sedimentation particles, it **still has significant pore density, which varies with the moisture content of the soil. So, statement III is not correct.**

So, only one of the Statements II and III is correct and that explains Statement I.

Therefore, option (c) is the correct answer.

Q29. Consider the following statements:

- Without the atmosphere, temperature would be well below freezing point everywhere on the Earth's surface.
 - Heat absorbed and trapped by the atmosphere maintains our planet's -average temperature.
 - Atmosphere's gases, like carbon dioxide, are particularly good at absorbing and trapping radiation.
- Which of the statements given above are correct?

- a) I and III only
- b) I and II only
- c) I, II and III
- d) II and III only

Answer: d

Explanation:

- The **atmosphere acts like a blanket of air around the earth**. This implies that the atmosphere holds in the amount of heat needed to keep the earth liveable (**Goldilocks' planet**).
- In the atmosphere certain gasses like carbon dioxide, water vapour etc. are known to absorb the infrared radiation of the sun, thus preventing the temperature on the surface of earth to plummet in the absence of sun, that is during the night time.
- Another way atmosphere maintains the temperature is by regulating the amounts of solar energy reaching the Earth, that is why the areas which has thicker layers of atmosphere (poles) receive lower solar energy, despite receiving equivalent radiation as the equatorial regions.
- In the absence of atmosphere, **the temperatures of the Earth's surface will exhibit marked variations**, reaching up to 100 degrees Celsius during the day, while plummeting to near freezing in the night, as is seen on the surface of the moon. **This freezing effect would be more pronounced on the regions where the sun rays are incident at an angle** (that is on poles and temperate regions) while the equatorial regions may have temperatures above the freezing point, as seen for the north pole and the south pole of the moon. **So, statement I is not correct.**
- The **greenhouse effect is the process through which heat is trapped near Earth's surface by certain gases**, helping to maintain a warmer temperature than it would have otherwise. **So, statement II is correct.**
- Greenhouse gases consist of **carbon dioxide, methane, ozone, nitrous oxide, chlorofluorocarbons and water vapour**. **So, statement III is correct.**

Therefore, option (d) is the correct answer.

Q30. Consider the following statements about the Rashtriya Gokul Mission:

- I. It is important for the upliftment of rural poor as majority of low producing indigenous animals are with small and marginal farmers and landless labourers.
- II. It was initiated to promote indigenous cattle and buffalo rearing and conservation in a scientific and holistic manner.

Which of the statements given above is/are correct?

- a) I only
- b) II only
- c) Both I and II
- d) Neither I nor II

Answer: c

Explanation:

- **Rashtriya Gokul Mission** was launched in December 2014 and is being implemented by the Department of Animal Husbandry and Dairying. It aims for the development and conservation of indigenous cattle and buffalo breeds, genetic upgradation of bovine population and enhancement of milk production and productivity of bovines.

- The scheme is important for the **upliftment of the rural poor as more than 80 per cent low low-producing indigenous animals are with small and marginal farmers and landless labourers. So, statement I is correct.**
- It strives for the **development and conservation of indigenous bovine breeds in a scientific, holistic manner. So, statement II is correct.**

Therefore, option (c) is the correct answer.

Q31. Consider the following statements:

Statement I:

Studies indicate that carbon dioxide emissions from cement industry account for more than 5% of global carbon emissions.

Statement II:

Silica-bearing clay is mixed with limestone while manufacturing cement.

Statement III :

Limestone is converted into lime during clinker production for cement manufacturing.

Which one of the following is correct in respect of the above statements?

- Both Statement II and Statement III are correct and both of them explain Statement I
- Both Statement II and Statement III are correct but only one of them explains Statement I
- Only one of the Statements II and III is correct and that explains Statement I
- Neither Statement II nor Statement III is correct

Answer: b

Explanation:

- Cement is an important construction ingredient produced in virtually all countries, it is primarily composed of a mixture of calcium carbonate mixed with silica, processed for making the mixture to undergo an exothermic reaction on being mixed with water at the time of application.
- As per a recent report by the World Economic Forum(WEF), **global cement manufacturing is responsible for about 8% of the world's total carbon dioxide emissions. So, statement I is correct.**
- Cement manufacturing is a complex process that **begins with mining and then grinding raw materials that include limestone and clay, to a fine powder, called raw meal.** This is a largely mechanical process where no significant emissions occur. **So, statement 2 is correct.**
- This raw meal is then heated to a sintering temperature as high as 1450 C in a kiln. In this process, **the chemical bonds of the raw materials are broken down, and the limestone (calcium carbonate) gets converted to lime (calcium oxide). The resultant product of the process is called clinker. So, statement 3 is correct.**
- Majority of the emissions from the cement manufacturing process occurs at this stage of clinker formation, both from decarbonisation of Limestone as well as the fuel emissions by the kiln.
- As per the WEF report, **unlike other sectors, where the largest share of emissions is energy-related, in cement and concrete production, more than half are process emissions.**

So, both Statement II and Statement III are correct but only one of them explains Statement I. Therefore, option (b) is the correct answer.

Q32. Consider the following statements:

Statement I:

At the 28th United Nations Climate Change Conference (COP28), India refrained from signing the 'Declaration on Climate and Health'.

Statement II:

The COP28 Declaration on Climate and Health is a binding declaration; and if signed, it becomes mandatory to decarbonize health sector.

Statement III :

If India's health sector is decarbonized, the resilience of its healthcare system may be compromised.

Which one of the following is correct in respect of the above statements?

- Both Statement II and Statement III are correct and both of them explain Statement I
- Both Statement II and Statement III are correct but only one of them explains Statement I
- Only one of the Statements II and III is correct and that explains Statement I
- Neither Statement II nor Statement III is correct

Answer: c

Explanation:

- During the COP28 of the United Nations Climate Change Conference, a total of 123 countries signed the **Declaration on Climate and Health**. Under this declaration the **signatory countries committed to incorporating health considerations in the context of the Paris Agreement and United Nations Framework Convention on Climate Change (UNFCCC) processes**. So, **statement I is correct**.
- The COP28 Declaration on Climate and Health is a **non-legally binding**, non-negotiated call to action and collective commitment that reflects the concerns and common position of countries on the importance of health within climate discourse and in the COP process. **So, statement 2 is not correct**.
- India refrained from signing this document as it believed that **going towards decarbonisation will require an overhaul of the already fragile and stretched healthcare sector of the country**, jeopardizing the health prospects of its vulnerable citizens. **So, statement 3 is correct**.

So, only one of the Statements II and III is correct and that explains Statement I.

Therefore, option (c) is the correct answer.

Q33. Consider the following statements:

Statement I:

Scientific studies suggest that a shift is taking place in the Earth's rotation and axis.

Statement II:

Solar flares and associated coronal mass ejections bombarded the Earth's outermost atmosphere with a tremendous amount of energy.

Statement III:

As the Earth's polar ice melts, the water tends to move towards the equator.

Which one of the following is correct in respect of the above statements?

- Both Statement II and Statement III are correct and both of them explain Statement I
- Both Statement II and Statement III are correct but only one of them explains Statement I
- Only one of the Statements II and III is correct and that explains Statement I
- Neither Statement II nor Statement III is correct

Answer: b

Explanation:

- Recent researches have shown that in the last two decades, **Earth's tilt has shifted by 31.5 inches, at the same time the rotational speed of the Earth has been slowing down**, thus increasing the length of the day. **So, statement I is correct.**
- This shift is largely caused due to the redistribution of masses across the planet. This **redistribution largely occurs when ice sheets and glaciers melt more than they grow from snowfall** and when aquifers lose more groundwater than precipitation replenishes, leading to shifting of mass from the poles to the equator. This causes the planet to wobble as it spins and its axis to shift location — a phenomenon called **polar motion**. They also cause Earth's rotation to slow, measured by the lengthening of the day. **So, statement 3 is correct.**
- A **solar flare** is a tremendous explosion on the Sun that happens when energy stored in 'twisted' magnetic fields (usually above sunspots) is suddenly released, these flares are often associated with **Coronal Mass Ejections (CMEs)** where charged materials from the Sun's corona reaches the earth's atmosphere. On reaction with the ionosphere it can produce effects such as radio blackouts, power outages, and beautiful auroras. **So, statement 2 is correct.**
- Though associated with tremendous amounts of energy these ejections on reaching the Earth are not energetic enough to cause any significant change in the axial tilt or rotation of the planet.

So, Both Statement II and Statement III are correct but only one of them explains Statement I Therefore, option (b) is the correct answer.

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Q34. Consider the following statements:

Statement I:

Article 6 of the Paris Agreement on climate change is frequently discussed in global discussions on sustainable development and climate change.

Statement II:

Article 6 of the Paris Agreement on climate change sets out the principles of carbon markets.

Statement III:

Article 6 of the Paris Agreement on climate change intends to promote inter-country non-market strategies to reach their climate targets.

Which one of the following is correct in respect of the above statements?

- Both Statement II and Statement III are correct and both of them explain Statement I
- Both Statement II and Statement III are correct but only one of them explains Statement I
- Only one of the Statements II and III is correct and that explains Statement I
- Neither Statement II nor Statement III is correct

Answer: a

Explanation:

- Article 6** of the Paris Agreement allows countries to voluntarily cooperate with each other to achieve emission reduction targets set out in their **Nationally Determined Contributions (NDCs)**, making it one of the talked about aspects of the Climate negotiations, and a reason for being in news. **So, statement I is correct.**
- Article 6 deals with the provision, a country (or countries) will be able to transfer carbon credits earned from the reduction of GHG emissions to help one or more countries meet climate targets. It establishes a **mechanism for trading GHG emission reductions between countries** under the supervision of the Conference of Parties, through the establishment of **carbon markets**. **So, statement 2 is correct.**
- Along with providing the mechanism for inter-country transfer of carbon credits via market cooperation **sub-article 6.8 of the Article 6, provides opportunities for non-market-based cooperation for enhancing climate action. So, statement 3 is correct.**

So, Both Statement II and Statement III are correct and both of them explain Statement I
Therefore, option (a) is the correct answer.

Q35. Which one of the following launched the 'Nature Solutions Finance Hub for Asia and the Pacific'?

- The Asian Development Bank (ADB)
- The Asian Infrastructure Investment Bank (AIIB)
- The New Development Bank (NDB)
- The International Bank for Reconstruction and Development (IBRD)

Answer: a

Explanation:

- The **Asian Development Bank (ADB)** launched the initiative **Nature Solutions Finance Hub for Asia and the Pacific** at COP28 of the United Nations Climate Change Conference which aims to attract at least \$2 billion into investment programs that incorporate nature-based solutions, particularly focused on capital markets and other sources of private capital.
- Nature-based solutions are features incorporated into the design of projects—both traditional infrastructure and direct nature-conservation projects—that protect, manage, and restore natural ecosystems while still enabling a country or community's growth.

- The initiative aims to **provide integrated activities that will support both upstream awareness, policy, and capacity building for nature-based solutions projects**, and downstream innovative finance structuring in projects to directly attract funds via bonds, from banks, and others.

Therefore, option (a) is the correct answer.

Q36. With reference to 'Direct Air Capture', an emerging technology, which of the following statements is/are correct?

- It can be used as a way of carbon sequestration.
 - It can be a valuable approach for plastic production and in food processing.
 - In aviation, it can be a source of carbon for combining with hydrogen to create synthetic low-carbon fuel.
- Select the correct answer using the code given below.

- I and II only
- III only
- I, II and III
- None of the above statements is correct

Answer: c

Explanation:

- **Direct air capture (DAC)** technologies extract carbon dioxide directly from the atmosphere at any location, unlike carbon capture which is generally carried out at the point of emissions, such as a steel plant.
- This **carbon dioxide can then be permanently stored in deep geological formations** or used for a variety of applications, hence **contributing to carbon sequestration. So, statement 1 is correct.**
- **The captured carbon can then be used as a feedstock for various carbon-based materials.** It can be used in the production of cement, **plastics, food processing**, carbon fibers, and other industrial applications. This helps keep carbon stored in the infrastructure, preventing its release into the atmosphere for extended periods. **So, statement 2 is correct.**
- Aviation is the most energy and carbon intensive mode of transport and the most difficult to decarbonise. Greenhouse gas emissions from aviation comprise around 2 % of global totals. The **DAC strategies can help reduce carbon intensity of the sector in two ways**, first is where continued use of fossil jet fuel is offset by capturing and sequestering CO₂ from the atmosphere. The second is where **carbon dioxide captured from the atmosphere is used as a feedstock, in conjunction with hydrogen from electrolysis, to produce a synthetic jet fuel. So, statement 3 is correct.**

Therefore, option (c) is the correct answer.

Q37. Regarding Peacock tarantula (Gooty tarantula), consider the following statements:

- It is an omnivorous crustacean.
- Its natural habitat in India is only limited to some forest areas.
- In its natural habitat, it is an arboreal species.

Which of the statements given above is/are correct?

- I only
- I and III
- II only
- II and III

Answer: d

Explanation:

- **Peacock tarantula**, known for its shiny metallic appearance, is one of the **most elusive and sought after species in the world**. Their unique aesthetics have made it catch the fancy of pet collectors across the world, contributing to the dwindling population of the already endemic arachnid. The peacock tarantula is listed as **Critically endangered in the IUCN red list**.
- It is a **carnivorous arthropod** belonging to the family **Arachnida** (not crustacean). **So, statement 1 is not correct.**
- This **arachnid is native to the Eastern Ghats of India**, with a very restricted range of mature forests. **So, statement 2 is correct.**
- This tarantula **lives in the crevices and holes of trees**, hence arboreal. Studies show that this species is specifically reliant on old growth forests. These trees not only offer shelter and nesting sites but also create a complex network of microhabitats that support the tarantula's prey. **So, statement 3 is correct.**

Therefore, option (d) is the correct answer.

Q38. Consider the following statements:

- Carbon dioxide (CO₂) emissions in India are less than 0.5 t CO₂/capita.
- In terms of CO₂ emissions from fuel combustion, India ranks second in Asia-Pacific region.
- Electricity and heat producers are the largest sources of CO₂ emissions in India.

Which of the statements given above is/are correct?

- I and III only
- II only
- II and III only
- I, II and III

Answer: c

Explanation:

- The **per capita emissions** of carbon dioxide refers to the amount of carbon dioxide emitted on a whole by the country divided by the total population of the country. Per-capita energy-related CO₂ emissions tend to be higher in more economically-developed countries, but can also vary greatly depending on the structure of the economy and the energy system.
- **India's per capita emissions in 2022 reached the levels of 1.78 tonnes**, marking a 111% rise over the last two decades. Though on a higher scale, India's per capita emissions are still on the lower side when compared to other developed economies. **So, statement 1 is not correct.**
- In terms of carbon dioxide emissions from fuel combustion, **India ranks second (after China) in the Asia-Pacific region**, with 10K metric tonnes of carbon being released from fossil fuels. **So, statement 2 is correct.**
- Energy is responsible for the majority of climate change-causing greenhouse gas emissions, mostly from the burning of fossil fuels. **In India, electricity and heat production contributes to 53% of the overall carbon dioxide emissions. So, statement 3 is correct.**

Therefore, option (c) is the correct answer.

Q39. Consider the following pairs:

	Plant	Description
I.	Cassava	: Woody shrub
II.	Ginger	: Herb with pseudostem
III.	Malabar spinach	: Herbaceous climber

IV. Mint : Annual shrub

V. Papaya : Woody shrub

How many of the above pairs are correctly matched?

- a) Only two
- b) Only three
- c) Only four
- d) All the five

Answer: b

Explanation:

- **Cassava:** It is a **woody shrub** belonging to the family *Euphorbiaceae*. Cassava is an annual crop **native to South America**. It is now extensively cultivated throughout tropical and subtropical regions, mainly for its edible tubers as a source of carbohydrates. **So, pair 1 is correctly matched.**
- **Ginger:** It is derived from the **rhizomes of *Zingiber officinale*** Rosc., a plant belonging to family Zingiberaceae. It is a slender perennial herb, mostly grown as an annual, 30–100 cm tall with a robust rhizome borne horizontally just below the soil surface. **Pseudostem is a stem made of intertwining of leaves giving an appearance of stem.** Ginger exhibits such character, along with large trees such as Banana. **So, pair 2 is correctly matched.**
- **Malabar Spinach:** It is a **fast growing, perennial vine** native to the Western Ghats, Ceylon and Indonesia. It is a **herbaceous climber with edible leaves**, which unlike traditional spinach are more bitter in flavour. **So, pair 3 is correctly matched.**
- **Mint:** It is an evergreen **perennial (not annual) herb**, which on distillation yields essential oils containing a large variety of aroma-chemicals in varying composition. These oils and their aroma-chemicals in pure form command a large and world-wide demand in trade. **So, pair 4 is not correctly matched.**
- **Papaya:** Papaya is an economically important fruit crop that belongs to the Caricaceae family and is native to the Mesoamerica region in America. It is grown worldwide in tropical and subtropical areas as a **semi-woody herbaceous perennial plant**. **So, pair 5 is not correctly matched.**

So, only three of the above given pairs are correctly matched.

Therefore, option (b) is the correct answer.

Q40. With reference to the planet Earth, consider the following statements:

- I. Rain forests produce more oxygen than that produced by oceans.
 - II. Marine phytoplankton and photosynthetic bacteria produce about 50% of the world's oxygen.
 - III. Well-oxygenated contains several surface water folds of higher oxygen than that in atmospheric air.
- Which of the statements given above is/are correct?

- a) I and II
- b) II only
- c) I and III
- d) None of the above statements is correct

Answer: b

Explanation:

- Scientists estimate that **roughly half of the oxygen production on Earth comes from the ocean.** The majority of this production is from oceanic plankton — drifting plants, algae, and some bacteria that can photosynthesize. **So, statement 2 is correct.**

- One particular species, *Prochlorococcus*, is the smallest photosynthetic organism on Earth. But this little bacteria produces up to 20% of the oxygen in our entire biosphere. That's a **higher percentage than all of the tropical rainforests on land combined**. So, **statement 1 is not correct**.
- It's important to remember that although the ocean produces at least 50% of the oxygen on Earth, **roughly the same amount is consumed by marine life**. Like animals on land, marine animals use oxygen to breathe and both plants and animals use oxygen for cellular respiration. Oxygen is also consumed when dead plants and animals decay in the ocean. That is why despite contributing nearly half of the oxygen, the concentration of oxygen in surface seawater is lower than the atmospheric air. So, **statement 3 is not correct**.

Therefore, option (b) is the correct answer.

Q41. Consider the following types of vehicles:

- Full battery electric vehicles
- Hydrogen fuel cell vehicles
- Fuel cell-electric hybrid vehicles

How many of the above are considered as alternative powertrain vehicles?

- Only one
- Only two
- All the three
- None

Answer: c

Explanation:

- A powertrain is a **system of components that transfer energy from the engine to the wheels to make a vehicle move**. The major components of a powertrain include the engine, transmission, driveshaft, differential and axles.
- **Alternative powertrains** include propulsion systems that are not based exclusively on the popular internal combustion engine. This may include **fuel cell hybrids, full battery electrics, hydrogen fuel cells**, compressed air, and many other types. So, **points 1, 2 and 3 are correct**.

So, all three of the above are considered as alternative powertrain vehicles.

Therefore, option (c) is the correct answer.

Q42. With reference to Unmanned Aerial Vehicles (UAVs), consider the following statements:

- All types of UAVs can do vertical landing.
- All types of UAVs can do automated hovering.
- All types of UAVs can use battery only as a source of power supply.

How many of the statements given above are correct?

- Only one
- Only two
- All the three
- None

Answer: d

Explanation:

- An **Unmanned Aerial Vehicle (UAV)** is a power-driven crew-less air vehicle. It follows the basic principle of aerodynamics and **uses an electric battery or gas-powered engine to fly**. In comparison to manned aircraft, UAVs are cheaper to design and manufacture. UAVs have higher mobility,

safety, and flexibility than other manned aircraft. Some UAVs do not need any specified airfield to take-off and land whereas for manned aircraft an airfield is mandatory.

- **Vertical landing** means, any air bound vehicle being capable of taking off or landing like a helicopter, that is generating a lift at zero speed. **Not all UAVs can manoeuvre a vertical landing. So, statement 1 is not correct.**
- **Automated hovering** means UAVs being able to maintain a stable position in the air without constant human intervention, relying on advanced sensors, algorithms to navigate and adjust their position. **Not all UAVs are capable of doing that. So, statement 2 is not correct.**
- **UAVs can be crafted on storage as well as combustion technology** based on the requirement of the host. **So, statement 3 is not correct.**

So, none of the above given statements are correct.

Therefore, option (d) is the correct answer.

Q43. In the context of electric vehicle batteries, consider the following elements:

- I. Cobalt
- II. Graphite
- III. Lithium
- IV. Nickel

How many of the above usually make up battery cathodes?

- a) Only one
- b) Only two
- c) Only three
- d) All the four

Answer: c

Explanation:

- **Cathodes are electrodes in a cell where the reduction takes place**, that is the components of the cathodes gain electrons to make the cell conduct electricity, on the contrary anode is the oxidising electrode where the loss of electrons usually take place.
- **Any positively charged ion can act as cathode for the purpose of electric vehicle batteries**, hence, metallic ions which have a propensity to form cations (positively charged ions).
- Therefore, metals like **Cobalt, Nickel and Lithium** can act as effective cathodes. **So, points 1, 3 and 4 are correct.**
- Graphite a single layered allotrope of carbon can function both as anode as well as cathode for the purpose of fuel cell battery, because it can entrap and lose electrons simultaneously, **but it is not usually used as cathode in conventional fuel batteries. So, point 2 is not correct.**

So, only three of the above usually make up battery cathodes.

Therefore, option (c) is the correct answer.

Q44. Consider the following:

- I. Cigarette butts
- II. Eyeglass lenses
- III. Car tyres

How many of them contain plastic?

- a) Only one
- b) Only two
- c) All the three

d) None

Answer: c

Explanation:

- Plastics are ubiquitous non-biodegradable compounds with multiple applications as structural as well as strengthening components.
- Plastics in cigarette butts are added to create a filtering component to prevent inhalation of aerosols and particulate matter. **So, point 1 is correct.**
- **Eyeglass lenses**, which were conventionally made of silicate glasses are now largely made of polycarbonate and poly acetate fibres (types of plastics) for their resilience and unbreakability. **So, point 2 is correct.**
- **Car tyres**, conventionally made of rubber and steel fibres, are nowadays being reinforced via plastic fibres to give extra resilience. **So, point 3 is correct.**

So, all three of the above given objects contain plastics.

Therefore, option (c) is the correct answer.

Q45. Consider the following substances:

I. Ethanol

II. Nitroglycerine

III. Urea

Coal gasification technology can be used in the production of how many of them?

- Only one
- Only two
- All three
- None

Answer: b

Explanation:

- **Coal gasification** is a **thermo-chemical process** that converts **coal into syngas**, a synthetic gas composed mainly of **carbon monoxide (CO)**, **hydrogen (H₂)**, **carbon dioxide (CO₂)**, **methane (CH₄)**, and **water vapour (H₂O)**.
- This synthetic gas can then be used in the manufacture of numerous chemicals both as direct feedstock and as fuel element.
- **Ethanol** is manufactured from syngas via both chemical and biological processes. For chemical synthesis, ethanol can be produced using catalyst-based processes that use methanol as the feedstock. **So, point 1 is correct.**
- **Nitroglycerin** is synthesized by nitrating glycerol with nitric acid, syngas is not directly involved in the production of this component either as feedstock or fuel. **So, point 2 is not correct.**
- In the manufacture of **urea syngas is the primary carbon donor**. **So, point 3 is correct.**

So, coal gasification technology is used in only two of the above given products.

Therefore, option (b) is the correct answer.

Q46. What is the common characteristic of the chemical substances generally known as CL-20, HMX and LLM-105, which are sometimes talked about in media?

- These are alternatives to hydro-fluorocarbon refrigerants

- b) These are explosives in military weapons
- c) These are high-energy fuels for cruise missiles
- d) These are fuels for rocket propulsion

Answer: b

Explanation:

- **CL-20:** It is chemically known as Hexanitrohexaazaisowurtzitane, it is a high-energy density material (HEDM) used in solid propellants and explosive mixtures. It's known for its high energy and density, making it a powerful explosive.
- **HMX:** It is an acronym for High Melting explosive, is also known as octogen and cyclotetramethylenetetranitramine, as well as by other names. It is a colorless solid that dissolves slightly in water. HMX is used in nuclear devices, plastic explosives, rocket fuels, and burster chargers. A small amount of HMX is also formed in making cyclotrimethylene-trinitramine (RDX), another explosive similar in structure to HMX.
- **LLM-105:** It is chemically known as 2,6-diamino-3,5-dinitropyrazine-I-oxide, is a high-performance, insensitive explosive material.

So, the above given chemical substances are used as explosives in military weapons.

Therefore, option (b) is the correct answer.

Q47. Consider the following statements:

- I. It is expected that Majorana I chip will enable quantum computing.
- II. Majorana I chip has been introduced by Amazon Web Services (AWS).
- III. Deep learning is a subset of machine learning.

Which of the statements given above are correct?

- a) I and II only
- b) II and III only
- c) I and III only
- d) I, II and III

Answer: c

Explanation:

- **Majorana I chip** is a quantum computing processor that utilizes a Topological Core architecture, enabling the creation of a more stable and scalable quantum processor.
- It's the first quantum chip to use a "Topoconductor" (Topological Superconductor), made of indium arsenide and aluminum, which allows for the creation of fermions, a unique type of particle that is both a particle and its own antiparticle.
- This architecture promises to **overcome the limitations of traditional qubit technologies**, paving the way for quantum computers capable of solving complex problems in various fields. **So, statement I is correct.**
- This chip is designed by the **Microsoft Corporation**. **So, statement 2 is not correct.**
- **Deep learning is a subset of machine learning** that uses multilayered neural networks, called deep neural networks, to simulate the complex decision-making power of the human brain. **So, statement 3 is correct.**

Therefore, option (c) is the correct answer.

Q48. With reference to monoclonal antibodies, often mentioned in news, consider the following statements:

- I. They are man-made proteins.
- II. They stimulate immunological function due to their ability to bind to specific antigens.
- III. They are used in treating viral infections like that of Nipah virus.

Which of the statements given above are correct?

- a) I and II only
- b) II and III only
- c) I and III only
- d) I, II and III

Answer: d

Explanation:

- **Monoclonal antibodies** are proteins made in laboratories that act like proteins called **antibodies** in human bodies. These are immunoglobulins derived from a monoclonal cell line and which have a defined specificity. **So, statement 1 is correct.**
- **Their immunological activities are based on binding to a specific ligand or antigen. So, statement 2 is correct.**
- Monoclonal antibodies have been used for **passive immunization (injection of preformed antibodies) therapy in treatment of Nipah virus infection. So, statement 3 is correct.**

Therefore, option (d) is the correct answer.

Q49. Consider the following statements:

I. No virus can survive in ocean waters.

II. No virus can infect bacteria.

III. No virus can change the cellular transcriptional activity in host cells.

How many of the statements given above are correct?

- a) Only one
- b) Only two
- c) All the three
- d) None

Answer: d

Explanation:

- Viruses are **obligate parasitic organisms** and are **ubiquitous in nature**, that is these organisms are found in almost all of the ecosystems of the earth.
- These parasites are considered living only when they are associated with a or are infecting a host, outside the host's body they are considered non-living as all kinds of metabolic activities cease in the virus at that time.
- **Viruses are abundantly found in the sea waters**, parasiting on marine animals like turtles, jellyfish etc. **So, statement 1 is not correct.**
- **Bacteriophage is a class of viruses which infect Bacterias. So, statement 2 is not correct.**
- Viruses belong to the **class retroviridae**, can take over the cellular machinery of the cell and can reverse the transcriptional activity of the host cell, to form DNA from RNA. **So, statement 3 is not correct.**

So, none of the above statements are correct.

Therefore, option (d) is the correct answer.

Q50. Consider the following statements:

Statement I:

Activated carbon is a good and an attractive tool to remove pollutants from effluent streams and to remediate contaminants from various industries.

Statement II:

Activated carbon exhibits a large surface area and a strong potential for adsorbing heavy metals.

Statement III:

Activated carbon can be easily synthesized from environmental wastes with high carbon content.

Which one of the following is correct in respect of the above statements?

- Both Statement II and Statement III are correct and both of them explain Statement I
- Both Statement II and Statement III are correct but only one of them explains Statement I
- Only one of the Statements II and III is correct and that explains Statement I
- Neither Statement II nor Statement III is correct

Answer: a

Explanation:

- Activated Carbons (ACs)** are a class of amorphous carbonaceous materials with large porosity and internal surface area.
- Since the **removal of contaminants requires carbonaceous materials with a high degree of porosity**, well-developed surface area, and distinct functional groups, use of ACs is one of the finest methods for eliminating pollutants from aqueous solution and the atmosphere. **So, statements I and 2 are correct.**
- Coconut shells, coal, and wood are the basic sources of ACs. Anthracite and bituminous coals have been the main sources of AC, **although AC may be made from any carbonaceous material, including environmental wastes**, adding to the cost effectiveness of the ACs. **So, statement 3 is correct.**

So, Both Statement II and Statement III are correct and both of them explain Statement I.

Therefore, option (a) is the correct answer.



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Q51. With reference to the Indian polity, consider the following statements:

- I. An Ordinance can amend any Central Act.
- II. An Ordinance can abridge a Fundamental Right.
- III. An Ordinance can come into effect from a back date.

Which of the statements given above are correct?

- a) I and II only
- b) II and III only
- c) I and III only
- d) I, II and III

Answer: c

Explanation:

- **Article 123** of the Constitution empowers the **President to promulgate ordinances during the recess of Parliament**. These ordinances have the same force and effect as an act of Parliament, but are in the nature of temporary law.
- His ordinance-making power is coextensive as regards all matters except duration, with the law-making powers of the Parliament. This has two implications:
 - An ordinance can be issued only on those subjects on which the Parliament can make laws.
 - An ordinance is subject to the same constitutional limitation as an act of Parliament. Hence, an ordinance **cannot abridge or take away any of the fundamental rights**. So, statement **II is not correct**.
- An ordinance like any other legislation, **can be retrospective, that is, it may come into force from a back date. It may modify or repeal any act of Parliament** or another ordinance. It can alter or amend a tax law also. However, it cannot be issued to amend the Constitution. So, statements **I and III are correct**.

Therefore, option (c) is the correct answer.

Q52. Consider the following pairs:

State	Description
I. Arunachal Pradesh	: The capital is named after a fort, and the State has two National Parks
II. Nagaland	: The State came into existence on the basis of a Constitutional Amendment Act
III Tripura	: Initially a Part 'C' State, it became a centrally administered territory with the reorganization of States in 1956 and later attained the status of a full-fledged State

How many of the above pairs are correctly matched?

- a) Only one
- b) Only two
- c) All the three
- d) None

Answer: b

Explanation:

- **Arunachal Pradesh:** The capital of the State is **Itanagar in Papumpare district**. Itanagar is named after **Ita fort meaning fort of bricks**, build in the 14th century AD. Arunachal Pradesh find mentioned in the literature of Kalika Purana and Mahabharata. Arunachal Pradesh has **two national parks, Namdapha National Park and Mouling National Park**. Namdapha National Park is one of the largest national parks in India and a biodiversity hotspot. Mouling National Park is located in the Upper Siang district. **So, pair I is correctly matched.**
- **Nagaland** attained statehood through the enactment of the **State of Nagaland Act, 1962**. Concurrently, the **Constitution (Thirteenth Amendment) Act, 1962** was introduced, incorporating Article 371A into the Constitution. This special provision accords Nagaland a unique status by safeguarding its distinctive religious beliefs, social customs, customary laws, and land ownership practices. It ensures that no Act of Parliament affecting these matters shall apply to the state unless approved by the Nagaland Legislative Assembly. **So, pair II is not correctly matched.**
- After death of the last king Maharaja Bir Bikram Kishore Manikya on May 17, 1947, a Regency Council was formed headed by Maharani Kanchan Prabha Devi, for aid of the minor Prince, Kirit Bikram Kishore Manikya Bahadur. The regent signed the merger agreement with the Government of India. After merger **Tripura became a Part 'C' State**. On reorganization of the states, effected in November, 1956, **Tripura became a Union Territory** with an Advisory Committee to aid and advise the Chief Commissioner. Thereafter in place of the Advisory Committee, a Territorial Council was formed through the adult franchise on 15th August, 1957. On July 1, 1963, the Tripura Territorial Council was dissolved and the Legislative Assembly with the existing members of the Territorial Council was formed. On January 21, 1972 **Tripura became full-fledged State by the Act of Parliament called the North Eastern Areas (Reorganization) Act, 1971**. **So, pair III is correctly matched.**

So, only two of the above pairs are correctly matched.

Therefore, option (b) is the correct answer.

Q53. With reference to India, consider the following:

- The Inter-State Council
- The National Security Council
- Zonal Councils

How many of the above were established as per the provisions of the Constitution of India?

- Only one
- Only two
- All the three
- None

Answer: a

Explanation:

- The '**Inter State Council**' is a **Constitutional body** established under **Article 263** of the Constitution of India. It was established on the recommendations of the Sarkaria Commission on Centre-State Relations (1983–88). It facilitates cooperation and coordination among States and the Central government. Thus, the **President can establish** such a council at any time it appears that the **public interest would be served by its establishment**. The President can define the nature of duties to be performed by such a council and its organisation and procedure. **So, point I is correct.**
- The **National Security Council (NSC) of India** is an apex body that advises the Prime Minister on national security and strategic interests. It operates under the **Executive Office of the Prime Minister**. The NSC is headed by the **National Security Advisor (NSA)** and includes various key officials, including the Chief of Defence Staff, the three service chiefs, and senior ministers. **So, point II is not correct.**

- **Zonal Councils** are **Statutory bodies** established under the **States Reorganisation Act, 1956**. This act divided the country into **five zones (Northern, Central, Eastern, Western and Southern)** and provided a Zonal Council for each zone. **So, point III is not correct.**

So, only one of the above were established as per the provisions of the Constitution of India. Therefore, option (a) is the correct answer.

Q54. Consider the following statements:

I. The Constitution of India explicitly mentions that in certain spheres the Governor of a State acts in his/her own discretion.

II. The President of India can, of his/her own, reserve a bill passed by a State Legislature for his/her consideration without it being forwarded by the Governor of the State concerned.

Which of the statements given above is/are correct?

- a) I only
- b) II only
- c) Both I and II
- d) Neither I nor II

Answer: a

Explanation:

- **Article 163–Council of Ministers to aid and advise Governor:** There shall be a Council of Ministers with the Chief Minister as the head to aid and advise the Governor in the exercise of his functions, except in so far as he is required to exercise his functions in his discretion. **So, statement I is correct.**
- When a bill is sent to the Governor after it is passed by State Legislature, he can:
 - Give his assent to the bill, or
 - Withhold his assent to the bill, or
 - Return the bill (if it is not a money bill) for reconsideration of the State Legislature. However, if the bill is passed again by the State Legislature with or without amendments, the Governor has to give his assent to the bill, or
 - Reserve the bill for the consideration of the President. In one case such reservation is obligatory, that is, where the bill passed by the State Legislature endangers the position of the state High Court. **President doesn't reserve independently. So, statement II is not correct.**

Therefore, option (a) is the correct answer.

Q55. Consider the following pairs:

Provision in the Constitution of India	Stated under
I. Separation of Judiciary from the Executive in the public services of the State	The Directive Principles of the State Policy
II. Valuing and preserving of the rich heritage of our composite culture	The Fundamental Duties
III. Prohibition of employment of children below the age of 14 years in factories	The Fundamental Rights

How many of the above pairs are correctly matched?

- a) Only one

- b) Only two
- c) All the three
- d) None

Answer: c

Explanation:

- The Constitution directs the state to take steps to separate the judiciary from the executive in public services. This means that the executive authorities should not possess the judicial powers. **To separate the judiciary from the executive in the public services of the State** (Article 50) is stated under the **Directive Principles of the State Policy**. So, pair I is correctly matched.
- According to **Article 51A of the Constitution of India**, it shall be the **duty of every citizen of India to value and preserve the rich heritage of the country's composite culture**. So, pair II is correctly matched.
- **Article 24 (Prohibition of employment of children in factories, etc.)** of the Constitution of India states that **no child below the age fourteen years shall be employed** in work in any factory or mine or engaged in any other hazardous employment. So, pair III is correctly matched.

So, all three of the above pairs are correctly matched.

Therefore, option (c) is the correct answer.

Q56. Consider the following statements:

With reference to the Constitution of India, if an area in a State is declared as Scheduled Area under the Fifth Schedule

- I. the State Government loses its executive power in such areas and a local body assumes total administration
- II. the Union Government can take over the total administration of such areas under certain circumstances on the recommendations of the Governor

Which of the statements given above is/are correct?

- a) I only
- b) II only
- c) Both I and II
- d) Neither I nor II

Answer: b

Explanation:

- The **Fifth Schedule of the Constitution** deals with the administration and control of scheduled areas and scheduled tribes in any state except the four states of Assam, Meghalaya, Tripura and Mizoram.
- The various features of administration contained in the Fifth Schedule are as follows:
 - The **executive power of a State extends to the scheduled areas therein**. But the Governor has a special responsibility regarding such areas. He has to submit a report to the President regarding the administration of such areas, annually or whenever so required by the President. The executive power of the Centre extends to giving directions to the States regarding the administration of such areas. **So, statement I is not correct.**
 - The President is empowered to declare an area to be a scheduled area. He can also increase or decrease its area, alter its boundary lines, rescind such designation or make fresh orders for such redesignation on an area in consultation with the Governor of the state concerned. The **Union**

Government can take over the total administration of such areas under certain circumstances on the recommendations of the Governor. **So, statement II is correct.**

Therefore, option (b) is the correct answer.

Q57. With reference to India, consider the following pairs :

Organization	Union Ministry
I. The National Automotive Board	: Ministry of Commerce and Industry
II. The Coir Board	: Ministry of Heavy Industries
III. The National Centre for Trade Information	: Ministry of Micro, Small and Medium Enterprises

How many of the above pairs are correctly matched?

- a) Only one
- b) Only two
- c) All the three
- d) None

Answer: d

Explanation:

- The **National Automotive Board (NAB)** falls under the **Ministry of Heavy Industries, Government of India**. It's an autonomous body incorporated as a **Society under the Societies Registration Act, 1860**. The Ministry of Heavy Industries oversees the NAB and its operations, including the implementation of NATRIP (National Automotive Testing & Research Program). **So, pair I is not correctly matched.**
- Coir Board is under the **Ministry of Micro, Small and Medium Enterprises (MSME), Govt. of India**. The Head Quarters of the Board is located at Coir House, M.G. Road, Kochi, Kerala and is running 48 establishments including 28 marketing outlets across the country. **So, pair II is not correctly matched.**
- The National Centre for Trade Information (NCTI) has been setup by Ministry of Commerce and Industry to create institutional mechanism for collection and dissemination of trade data and for improvement of business information services. **So, pair III is not correctly matched.**

So, none of the above pairs are correctly matched.

Therefore, option (d) is the correct answer.

Q58. Consider the following subjects under the Constitution of India:

- I. List I-Union List, in the Seventh Schedule
- II. Extent of the executive power of a State
- III. Conditions of the Governor's office

For a constitutional amendment with respect to which of the above, ratification by the Legislatures of not less than one-half of the States is required before presenting the bill to the President of India for assent?

- a) I and II only
- b) II and III only
- c) I and III only
- d) I, II and III

Answer: a

Explanation:

- **Article 368 of Constitution of India** provides for two types of amendments, that is, by a **special majority of Parliament** and also through the **ratification of half of the states by a simple**

majority. But, some other articles provide for the amendment of certain provisions of the Constitution by a simple majority of Parliament, that is, a majority of the members of each House present and voting (similar to the ordinary legislative process).

- Those provisions of the Constitution which are related to the **federal structure of the polity can be amended by a special majority of the Parliament** and also with the **consent of half of the State Legislatures by a simple majority.** If one or some or all the remaining states take no action on the bill, it does not matter; the moment half of the states give their consent, the formality is completed. There is no time limit within which the states should give their consent to the bill.
- The following provisions can be amended in this way:
 - Election of the President and its manner.
 - **Extent of the executive power of the Union and the states. So, point II is correct.**
 - Supreme Court and High Courts.
 - Distribution of legislative powers between the Union and the states.
 - Goods and Services Tax Council.
 - **Any of the lists in the Seventh Schedule. So, point I is correct.**
 - Representation of states in Parliament.
 - Power of Parliament to amend the Constitution and its procedure (Article 368 itself).
- **Conditions of the Governor's office** can be amended without the ratification by the Legislatures of not less than one-half of the States. **So, point III is not correct.**

Therefore, option (a) is the correct answer.

Q59. With reference to the Indian polity, consider the following statements:

I. The Governor of a State is not answerable to any court for the exercise and performance of the powers and duties of his/her office.

II. No criminal proceedings shall be instituted or continued against the Governor during his/her term of office.

III. Members of a State Legislature are not liable to any proceedings in any court in respect of anything said within the House.

Which of the statements given above are correct?

- a) I and II only
- b) II and III only
- c) I and III only
- d) I, II and III

Answer: d

Explanation:

- According to the **Article 361 in Constitution of India**, the **President, or the Governor of a State**, shall not be **answerable to any court for the exercise and performance** of the powers and duties of his office or for any act done or purporting to be done by him in the exercise and performance of those powers and duties. **So, statement I is correct.**
- **No criminal proceedings** can be started against the **President and the Governors** in respect of their personal acts **nor can they be arrested or imprisoned.** This immunity is limited to the period of the term of their office only and does not extend beyond that. However, civil proceedings can be started against them during their term of office in respect of their personal acts after giving two months' advance notice. **So, statement II is correct.**

- Members of a State Legislature have freedom of speech in the State Legislature. **No member is liable to any proceedings in any court for anything said or any vote given by him in the State Legislature or its committees.** This freedom is subject to the provisions of the Constitution and to the rules and standing orders regulating the procedure of the State Legislature. **So, statement III is correct.**

Therefore, option (d) is the correct answer.

Q60. Consider the following activities:

- Production of crude oil
- Refining, storage and distribution of petroleum
- Marketing and sale of petroleum products
- Production of natural gas

How many of the above activities are regulated by the Petroleum and Natural Gas Regulatory Board in our country?

- Only one
- Only two
- Only three
- All the four

Answer: b

Explanation:

- The **Petroleum and Natural Gas Regulatory Board (PNGRB)** was established under the **Petroleum and Natural Gas Regulatory Board Act, 2006**. According to the Act, the PNGRB is mandated to regulate the following activities:
 - Refining, processing, storage, transportation, distribution, marketing and sale of petroleum, petroleum products and natural gas. **So, points II and III are correct.**
 - However, the Act explicitly excludes the **production of crude oil and natural gas** from the purview of PNGRB's regulatory functions. **So, points I and IV are not correct.**

So, only two of the above activities are regulated by the Petroleum and Natural Gas Regulatory Board in our country.

Therefore, option (b) is the correct answer.

Q61. Suppose the revenue expenditure is ₹80,000 crores and the revenue receipts of the Government are 60,000 crores. The Government budget also shows borrowings of ₹10,000 crores and interest payments of 6,000 crores. Which of the following statements are correct?

- Revenue deficit is ₹20,000 crores.
- Fiscal deficit is ₹10,000 crores.
- Primary deficit is ₹4,000 crores.

Select the correct answer using the code given below:

- I and II only
- II and III only
- I and III only
- I, II and III

Answer: d

Explanation:

- Revenue Deficit** is the excess of revenue expenditure over revenue receipts.
 - Revenue Expenditure = ₹80,000 crores
 - Revenue Receipts = ₹60,000 crores

- Therefore, Revenue Deficit becomes (₹80,000 - ₹60,000) that is ₹20,000 crores. **So, statement I is correct.**
- Fiscal Deficit is the total borrowing requirement of the government and is calculated as:
 - Fiscal Deficit = Total Expenditure - Total Receipts (excluding borrowings)
 - Since borrowings are ₹10,000 crores (as given), this itself represents the fiscal deficit. **So, statement II is correct.**
- Primary Deficit is taken to as Fiscal Deficit minus the Interest Payments that is ₹10,000 crores minus ₹6,000 crores, hence it becomes ₹4,000 crores. **So, statement III is correct.**

Therefore, option (d) is the correct answer

Q62. India is one of the founding members of the International North-South Transport Corridor (INSTC), a multimodal transportation corridor, which will connect

- a) India to Central Asia to Europe via Iran
- b) India to Central Asia via China
- c) India to South-East Asia through Bangladesh and Myanmar
- d) India to Europe through Azerbaijan

Answer: a

Explanation:

- The **International North-South Transport Corridor (INSTC)** is an important initiative taken by India, Russia and Iran by signing an Inter-Governmental Agreement on 12th September 2000, to enhance trade and transport connectivity among countries along its route. **This corridor connects India Ocean and Persian Gulf to the Caspian Sea via Islamic republic of Iran**
- At present, there are 13 Members of INSTC, namely- India, Iran, Russia, Azerbaijan, Armenia, Kazakhstan, Kyrgyzstan, Tajikistan, Turkey, Ukraine, Belarus, Oman and Syria. Bulgaria has joined as an Observer State.
- **It has the potential to enhance India's connectivity with Central Asia and Eurasian Region and vice versa**, taking into account the geo-strategic and economic importance of all the involved countries.

Therefore, option (a) is the correct answer

Q63. Consider the following statements:

Statement I:

Of the two major ethanol producers in the world, i.e., Brazil and the United States of America, the former produces more ethanol than the latter.

Statement II:

Unlike in the United States of America where corn is the principal feedstock for ethanol production, sugarcane is the principal feedstock for ethanol production in Brazil.

Which one of the following is correct in respect of the above statements?

- a) Both Statement I and Statement II are correct and Statement explains Statement I II
- b) Both Statement I and Statement II are correct but Statement II does **not** explain Statement I
- c) Statement I is correct but Statement II is not correct
- d) Statement I is not correct but Statement II is correct

Answer: d

Explanation:

- **Ethanol is formed from the fermentation of sugars** in their natural occurrences or being derived from starch-rich grains or lignocellulosic feedstock, with sugar cane and corn being used as the major feedstock for ethanol production.
- Ethanol produced from various bio-based sources (bioethanol) has been gaining high attention lately due to its potential to cut down net emissions of carbon dioxide while reducing burgeoning world dependence on fossil fuels
- Amongst countries, **United States of America (USA) and Brazil are the topmost producers**, with the **USA producing nearly 15 gallons while Brazil contribution 7 gallons. So, statement I is not correct.**
- The vast majority of **U.S. ethanol is produced from corn, while Brazil primarily uses sugarcane. So, statement II is correct.**

**So, statement I is not correct but Statement II is correct
Therefore, option (d) is the correct answer.**

Q64. The World Bank warned that India could become one of the first places where wet-bulb temperatures routinely exceed 35 °C. Which of the following statements best reflect(s) the implication of the above-said report?

I. Peninsular India will most likely suffer from flooding, tropical cyclones and droughts.

II. The survival of animals including humans will be affected as shedding of their body heat through perspiration becomes difficult.

Select the correct answer using the code given below:

- I only
- II only
- Both I and II
- Neither I nor II

Answer: c

Explanation:

- The **Wet-bulb temperature (WBT)** is the lowest temperature achieved by adiabatic evaporation (= without adding heat or with equal enthalpy) of water in the air until the air is saturated (absolute humidity rises).
- Heat is required to evaporate water. If this evaporation is adiabatic, the heat is extracted from the environment (e.g. the air), causing the air to cool down and become more humid, this process will continue to do so until the air around the system becomes saturated with water or reach the levels of absolute humidity. With increasing levels of humidity this cooling effect of evaporation gets diminished/weakened. Thus, WBT can be taken to as a measure of effect of humidity on animals and environment.
- In **animals sweating is a physiological process in response to heat stress on the body**, the sweat by means of evaporation from the skin absorbs the internal heat, leading to maintenance of body temperature. In situations of increased humidity, that is increased wet bulb temperatures, this cooling effect of sweat is lost, thus accentuating the heat stress.
- **World Bank in its Climate Risk Country Profile for India have cited that researches have placed a threshold of 35°C (wet bulb ambient air temperature) on the human body's ability to regulate temperature**, beyond which even a very short period of exposure can present risk of serious ill-health and death. **So, statement II is correct.**

- The impact is likely to include **increased rainfall during the monsoon season**, which could heighten the risk of flooding across the Indian peninsula. Additionally, the region faces a growing threat of more frequent **tropical cyclones and prolonged droughts**. So, **statement I is correct**.

Therefore, option (c) is the correct answer.

Q65. A country's fiscal deficit stands at ₹50,000 crores. It is receiving ₹10,000 crores through non-debt creating capital receipts. The country's interest liabilities are ₹1,500 crores. What is the gross primary deficit?

- a) ₹48,500 crores
- b) ₹51,500 crores
- c) ₹58,500 crores
- d) None of the above

Answer: a

Explanation:

- In public finance**, deficits represent the shortfall between the government's revenue and its expenditure. Various types of deficits are used to evaluate the fiscal health of the government. These deficits are essential for **understanding how the government finances its expenditures**, borrows money, and the implications for the economy. Below are the most commonly referred to deficits.
- Fiscal Deficit (FD)** is the gap between a government's total expenditure and its total non-borrowed revenue (like taxes and fees).

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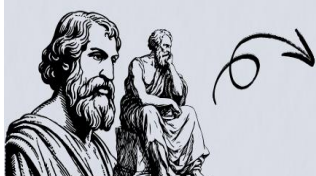
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- **Primary Deficit (PD)** is the difference between what a government earns (not counting interest payments) and what it spends on non-interest items.
- Hence, primary deficit essentially becomes a mathematical expression of fiscal deficit excluding interest payments.

$$\text{PD} = \text{FD} - \text{interest payments} \quad \dots\dots(I)$$

- In the above scenario, ₹50,000 Cr. is the fiscal deficit and ₹1500 Cr. is the interest liabilities, therefore to calculate primary deficit in this scenario we have to subtract these two figures.
- Putting values in equation (I)

$$\text{PD} = (50,000\text{Cr.}) - (1500\text{Cr.})$$

$$\text{PD} = 48,500\text{Cr.}$$

So, the Primary Deficit becomes ₹48,500Cr.

Therefore, option (a) is the correct answer.

Q66. Which of the following statements with regard to recommendations of the 15th Finance Commission of India are correct?

- It has recommended grants of 4,800 crores from the year 2022-23 to the year 2025-26 for incentivizing States to enhance educational outcomes.
- 45% of the net proceeds of Union taxes are to be shared with States.
- 45,000 crores are to be kept as performance-based incentive for all States for carrying out agricultural reforms.
- It reintroduced tax effort criteria to reward fiscal performance.

Select the correct answer using the code given below.

- I, II and III
- I, II and IV
- I, III and IV
- II, III and IV

Answer: c

Explanation:

- The **Finance Commission is a constitutional body under Article 280 of the Constitution of India** formed by the President of India to give suggestions on centre-state financial relations.
- The 15th finance commission was tasked for providing recommendations for financial devolution between Centre and States for the five-year period between FY2021 to FY2026.
- The share of states in the central taxes for the 2021-26 period was **recommended to be 41%**. This was less than the 42% share recommended by the 14th Finance Commission for 2015-20 period. The adjustment of 1% was to provide for the newly formed Union Territories of Jammu and Kashmir and Ladakh from the resources of the Centre. **So, statement II is not correct.**
- The Fifteenth Finance Commission's terms of reference was unique and wide ranging in many ways. Unlike the earlier commissions which were tasked for merely recommendations for the vertical and horizontal devolution of revenues amongst the Centre and the States, the fifteenth finance commission was asked to recommend performance incentives for States in many areas like power sector, adoption of Direct Benefit Transfers, solid waste management etc. as well.
- **The 15th Finance Commission recommended grants of Rs. 4,800 crore (Rs. 1,200 crore each year) from 2022-23 to 2025-26** for incentivising the States to enhance educational outcomes. **So, statement I is correct.**

- The commission further **recommended that Rs. 45,000 crore should be kept as performance-based incentive for all the States for carrying out agricultural reforms**, such as
 - amending their land-related laws on the lines of NITI Aayog's model law
 - maintaining and augmenting groundwater stock.
 - growth in agricultural exports
 - production of oilseeds, pulses and wood and wood-based products. **So, statement III is correct.**
- The Finance Commission adopted the following criteria for determining the share of devolution to the states:

Criteria	Percentage
Income Distance	45.0
Area	15.0
Population (2011) [#]	15.0
Demographic Performance	12.5
Forest and Ecology	10.0
Tax efforts	2.5

- **Tax efforts** for the purpose of devolution was defined as the ratio of the average per capita own tax revenue and the average per capita state GDP during the three years between 2016-17 and 2018-19. This criterion has been used to reward states with higher tax collection efficiency. **So, statement IV is correct.**

Therefore, option (c) is the correct answer.

Q67. Consider the following statements in respect of the International Bank for Reconstruction (IBRD) and Development:

- I. It provides loans and guarantees to middle income countries.
- II. It works single-handedly to help developing countries to reduce poverty.
- III. It was established to help Europe rebuild after the World War II.

Which of the statements given above are correct?

- g) I and II only
- b) II and III only
- c) I and III only
- d) I, II and III

Answer: c

Explanation:

- The **International Bank for Reconstruction and Development (IBRD)** is the oldest amongst the five World Bank institutions.
- This **institution was initially formed in 1945 to function in the area of the reconstruction of the war ravaged regions from World War II. So, statement III is correct.**
- Later it also assumed the function of **providing development assistance via loans to the middle income countries as well as credit worthy poorer countries. So, statement I is correct.**

- IBRD works closely with all institutions of the World Bank Group and the public and private sectors in developing countries to reduce poverty and build shared prosperity. **So, statement II is not correct.**

Therefore, option (c) is the correct answer.

Q68. Consider the following statements in respect of RTGS and NEFT:

- In RTGS, the settlement time is instantaneous while in case of NEFT, it takes some time to settle payments.
- In RTGS, the customer is charged for inward transactions while that is not the case for NEFT.
- Operating hours for RTGS are restricted on certain days while this is not true for NEFT.

Which of the statements given above is/are correct?

- I only
- I and II
- I and III
- III only

Answer: a

Explanation:

- **National Electronic Fund Transfer (NEFT) and Real Time Gross Settlement (RTGS)** are two electronic payment systems in India that enable the transfer of money between bank accounts.
- **RTGS** is a system where there is continuous and **real-time settlement of fund-transfers, individually on a transaction-by-transaction basis (without netting)**. 'Real Time' means the processing of instructions at the time they are received; 'Gross Settlement' means that the settlement of funds transfer instructions occurs individually.
- National Electronic Funds Transfer (NEFT) is a nation-wide centralised payment system owned and operated by the Reserve Bank of India (RBI).
- **NEFT is an electronic fund transfer system in which the transactions received up to a particular time are processed in batches (every two hours)**. Contrary to this, in RTGS, the transactions are processed continuously on a transaction-by-transaction basis throughout the day. **So, statement I is correct.**
- The RTGS system is primarily meant for large value transactions. The minimum amount to be remitted through RTGS is ₹2,00,000/- with no upper or maximum ceiling. While NEFT doesn't have any minimum limit.
- Inward transactions refer to the transactions received in the bank account of the receiver. It is to be noted here that, **in India for no payment system there is charges on inward transactions. So, statement II is not correct.**
- Both, **RTGS and NEFT are available for 24 hours of the day. So, statement III is not correct.**

Therefore, option (a) is the correct answer.

Q69. Consider the following countries:

- United Arab Emirates
- France
- Germany
- Singapore
- Bangladesh

How many countries amongst the above are there other than India where international merchant payments are accepted under UPI?

- a) Only two
- b) Only three
- c) Only four
- d) All the five

Answer: b

Explanation:

- **Unified Payments Interface (UPI)** is a real time digital payment system of India system that powers multiple bank accounts into a single mobile application (of any participating bank), merging several banking features and Peer to Peer collection request.
- As of recent updates, **UPI has been accepted for international merchant payments** in countries such as the **United Arab Emirates, Singapore and Bangladesh**. These countries have agreements or arrangements allowing UPI-based payments for cross-border transactions. **So, points I, IV and V are correct.**
- **France and Germany**, as of now, do not officially support international merchant payments through UPI. **So, points II and III are not correct.**

So, only three of the above given countries support international merchant payments under UPI. Therefore, option (b) is the correct answer.

Q70. Consider the following statements about 'PM Surya Ghar Free Electricity Scheme':

- I. It targets installation of one crore solar rooftop panels in the residential sector.
- II. The Ministry of New and Renewable Energy aims to impart training on installation, operation, maintenance and repairs of solar rooftop systems at grassroot levels.
- III. It aims to create more than three lakhs skilled manpower through fresh skilling, and up-skilling, under scheme component of capacity building.

Which of the statements given above are correct?

- a) I and II only
- b) I and III only
- c) II and III only
- d) I, II and III

Answer: d

Explanation:

- The Government of India launched the **PM Surya Ghar Muft Bijli Yojana** in February, 2024 to increase the share of solar rooftop capacity and empower residential households to generate their own electricity.
- It is a Central Sector Scheme with an objective **to install 1 crore rooftops in the residential sector**. **So, statement I is correct.**
- Under this scheme the **Ministry of New and Renewable Energy (MNRE)** intends to impart **training on installation, operation, maintenance and repairing of solar roof top systems at grass root level**. **So, statement II is correct.**
- It also aims to create **more than 3 lakh skilled manpower, through fresh skilling, and up-skilling**, under scheme component of capacity building, out of which at least 1,00,000 will be Solar PV Technicians. **So, statement III is correct.**

Therefore, option (d) is the correct answer.

Q71. "Sedition has become my religion" was the famous statement given by Gandhiji at the time of

- a) the Champaran Satyagraha

- b) publicly violating Salt Law at Dandi
- c) attending the Second Round Table Conference in London
- d) the launch of the Quit India Movement

Answer: b

Explanation:

- Gandhiji was accused of the offence of sedition multiple times in his political career, right from his days in South Africa to his **articles in Young India**, Gandhi's tryst with sedition law continued with full impunity, at times even he himself pleading guilty.
- This led to **Gandhi calling Section 124A of the Indian Penal Code which dealt with sedition as the prince among the political sections of the Indian Penal Code** designed to suppress the liberty of the citizen.
- His criticism of the law went one step ahead, **when on the eve of Dandi March, in 1930, he in his speech at Borsad quietly embracingly said 'Sedition has now become my religion'.**

Therefore, option (b) is the correct answer.

Q72. The famous female figurine known as 'Dancing Girl', found at Mohenjo-daro, is made of

- a) carnelian
- b) clay
- c) bronze
- d) gold

Answer: c

Explanation:

- The course of **Indus Valley Civilization (IVC)** exhibit the maturation of the civilization from **chalcolithic (copper) to bronze age civilization**, with the early sites exhibiting large number of copper artefacts, while the later stages were dominated by Bronze.
- Harappans were skilled in the arts of alloy making (Bronze is an alloy of copper), Alloys increased over time – for instance at Mohenjodaro, bronze tools increased from 6% to 23% from the lower to the higher levels.
- **The famed dancing girl of Mohenjodaro, discovered from the site in 1926-27 excavations is entirely made up of bronze casted by using the lost wax method.**

Therefore, option (c) is the correct answer.

Q73. Who provided legal defence to the people arrested in the aftermath of Chauri Chaura incident?

- a) C.R. Das
- b) Madan Mohan Malaviya and Krishna Kant
- c) Dr. Saifuddin Kitchlew and Khwaja Hasan Nizami
- d) M. A. Jinnah

Answer: b

Explanation:

- The **Chauri Chaura incident**, where a mob of protestors associated with the **Non Cooperation Movement (NCM)** in a fit of rage burnt down a police station in the Chauri Chaura area of the erstwhile United Province, leading to the death of 21 policemen.
- On one side it led to the withdrawal of the nationwide movement by Gandhi, on the other side the protestors who have now become accused were unable to defend for themselves in court, with congress already distancing itself from them.

- At this juncture **Pt. Madan Mohan Malviya** who had earlier retired from legal practice way back in 1911, came out of the ascetic life to defend the freedom fighters convicted in the case.
- **He along with Barrister Krishna Kant were able to secure acquisition of 156 accused out of 177.**

Therefore, option (b) is the correct answer.

Q74. Subsequent to which one of the following events, Gandhiji, who consistently opposed untouchability and appealed for its eradication from all spheres, decided to include the upliftment of 'Harijans' in his political and social programme?

- The Poona Pact
- The Gandhi-Irwin (Delhi Pact) Agreement
- Arrest of Congress leadership at the time of the Quit India Movement
- Promulgation of the Government of India Act, 1935

Answer: a

Explanation:

- **Poona pact** was an agreement between **MK Gandhi and BR Ambedkar at the Yerwada jail in Poona**, where the latter withdrew its demand for the separate electorates for Dalits in lieu of the reservation of seats for the Dalits in provincial and Central legislatures.
- During the negotiations of this pact Gandhi said to Ambedkar that **as a part of penance he has accepted himself to be a harijan, and from now on he will dedicate all his life for the upliftment of his community, that is the Harijan community.**
- Post the pact Gandhi retired from active politics and established the All India Untouchability league, for the upliftment of Harijans and appealed for eradication of untouchability from all spheres of the society.

Therefore, option (a) is the correct answer.

Q75. Consider the following fruits:

- Papaya
- Pineapple
- Guava

How many of the above were introduced in India by the Portuguese in the sixteenth and seventeenth centuries?

- Only one
- Only two
- All the three
- None

Answer: c

Explanation:

- Portuguese were the pioneer amongst Europeans to reach India via the sea route, in their colonial pursuits they have already asserted their dominance over other tropical countries in different continents.
- The presence of a common overlord led to increased interactions amongst these countries, with many crops and fruits of these regions finding inroads in geographical regions far away from their native regions.
- These crops because they were migrating amongst the similar geographic regions thrived in their new homes as well and soon became a part of the natives' palate and plate and sooner a part of the cultural milieu.
- Some crops introduced by the Portugese to India:
 - **Papaya:** It is a native of Central America, and was brought to India by Portugese traders. **So, point I is correct.**

- o **Pineapple:** It is a tropical citrus fruit native to tropical South America, particularly southern Brazil and Paraguay. It was also brought to India by the Portuguese. **So, point II is correct.**
- o **Guava:** It was brought to India from Latin America by the Portuguese. **So, point III is correct.**

So, all three of the above given fruits were introduced in India by the Portuguese in the sixteenth and seventeenth centuries.

Therefore, option (c) is the correct answer.

Q76. Consider the following countries:

- I. United Kingdom
- II. Denmark
- III. New Zealand
- IV. Australia
- V. Brazil

How many of the above countries have more than four time zones?

- a) All the five
- b) Only four
- c) Only three
- d) Only two

Answer: c

Explanation:

- A **time zone** is a standard time allocated to a region, approximately 15 degrees wide longitudinally. It is defined by the difference, or offset, from **Coordinated Universal Time (UTC)**, written as UTC- or UTC+ to indicate the hours ahead or behind UTC.
- Various countries adopt a standard meridian passing through their country to be taken as the reference time zone, for example **India has adopted 83.5°E as its prime meridian**, and accordingly the time across the length and breadth of the country is appropriated according to it.
- While India which extends from 68°7'E to 97°25'E longitude has adopted for a single time zone for whole of the country, some countries do adopt for multiple time zones within their territory.
- Of the given list of countries:
 - The **United Kingdom has one standard time zone. So, point I is not correct.**
 - **Denmark spans five time zones**, including Greenland, which operates at UTC-04:00, while the main territory operates at UTC+01:00. Greenland itself has three additional time zones, mostly at UTC-03:00. Near the Arctic Circle, time becomes fluid, with summers of perpetual daylight and winters of prolonged darkness, making time zones crucial for communication and balance. **So, point 2 is correct.**
 - **New Zealand has five time zones.** Between New Zealand's islands of Niue and Tokelau, the dividing line for UTC cuts somewhat arbitrarily. New Zealand's three other time zones are situated in its main territory, the Cook Islands and the Chatham Islands. **So, point 3 is correct.**
 - **Australia has nine time zones.** Australia's expansive territory includes its mainland and nearby islands. Settlement on the mainland dates back thousands of years, with British control established in 1788 through convict colonisation. Australia claims islands like **Heard and McDonald Islands (UTC+05:00)**, **Christmas Island (UTC+07:00)**, and **Norfolk Island (UTC+11:00)**, while its mainland operates within time zones ranging from UTC+08:00 to UTC+10:00. **So, point 4 is correct.**

- **Brazil uses four time zones.** Its easternmost zone is UTC-02:00, around the Rocas Atoll and Saint Peter and Saint Paul Archipelago. To the west, the legendary Amazonas territory is on UTC-05:00. **Brazil has four time zones. So, point 5 is not correct.**

So, only three of the above given countries have more than four time zones.

Therefore, option (c) is the correct answer.

Q77. Consider the following statements:

I. Anadyr in Siberia and Nome in Alaska are a few kilometers from each other, but when people are waking up and getting set for breakfast in these cities, it would be different days.

II. When it is Monday in Anadyr, it is Tuesday in Nome.

Which of the statements given above is/are correct?

- a) I only
- b) II only
- c) Both I and II
- d) Neither I nor II

Answer: a

Explanation:

- The **International Date Line (IDL)**, recognized in 1884, **passes through the mid-Pacific Ocean and roughly follows a 180 degrees longitude north-south line on the Earth.** It is located halfway around the world from the prime meridian — the 0 degrees longitude line in Greenwich, England.
- The international date line functions as a **line of demarcation** separating two consecutive calendar dates. **When one crosses the date line East to West it loses one day**, that is if a person crossing



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over from immediately east of the IDL on a Wednesday will reach the immediately West of the IDL on Thursday.

- The town of Anadyr in Russia, is located at approximately $177^{\circ}31'E$ longitude, while Nome, Alaska, USA, is located at $165^{\circ}23'58"W$ longitude, while being separated from each other by just a few nautical miles by the Bering strait.
- But, as these two cities are situated the just opposite sides of IDL, there is a difference in dates in the two, with time difference of being near to 24 hours. So, statement I is correct.
- Anadyr is located eastwards of the IDL, while Nome is westwards, keeping the earlier principle in mind (one gains a day while crossing the IDL from East to West). Hence, when it is Monday in Anadyr, people in Nome will still be enjoying their Sunday. So, statement II is not correct.

Therefore, option (a) is the correct answer.

Q78. Who among the following was the founder of the 'Self-Respect Movement'?

- Periyar' E.V. Ramaswamy Naicker
- Dr. B. R. Ambedkar
- Bhaskarrao Jadhav
- Dinkarrao Javalkar

Answer: a

Explanation:

- **Self-Respect Movement** was a social movement in **South India** aimed at achieving social equality and challenging the caste system. It was founded by **Periyar E.V. Ramaswamy Naicker**, along with S. Ramanathan in 1925. The movement aimed to challenge and dismantle the dominance of Brahminism and caste-based discrimination and promote self-respect among lower castes.
- It emphasized on the dignity and equality of all individuals, advocating for social justice and rational thinking. It significantly impacted the socio-political landscape of South India, leading to the rise of Dravidian ideologies and influencing subsequent political developments.

Therefore, option (a) is the correct answer.

Q79. Consider the following pairs:

Country	Resource-rich in
I. Botswana	: Diamond
II. Chile	: Lithium
III. Indonesia	: Nickel

In how many of the above rows is the given information correctly matched?

- Only one
- Only two
- All the three
- None

Answer: c

Explanation:

- **Botswana:** Botswana, a sparsely populated and arid country in **southern Africa**, is the **biggest producer of diamonds by value** and the second biggest by volume behind Russia. Diamonds account for around 80% of Botswana's exports and a quarter of its GDP. So, pair I is correctly matched.
- **Chile:** This South American country is the **world's third-largest holder of lithium resources**, after Bolivia and Argentina, and is the **largest holder of lithium reserves**, with 9.3 million tons. So, pair II is correctly matched.

- **Indonesia:** It has the **largest reserves of nickel in the world**. Until recently, it mostly sold its nickel deposits raw. It didn't have the processing plants to refine the metal. **So, pair III is correctly matched. Therefore, option (c) is the correct answer.**

Q80. Consider the following pairs:

Region	Country
I. Mallorca	: Italy
II. Normandy	: Spain
III. Sardinia	: France

In how many of the above rows is the given information correctly matched?

- Only one
- Only two
- All the three
- None

Answer: d

Explanation:

- **Mallorca:** It is an island of **Spain** in the Mediterranean Sea. **So, pair I is not correctly matched.**
- **Normandy:** It is a cultural region in the **French mainland**. **So, pair II is not correctly matched.**
- **Sardinia:** Sardinia is the second largest island in the Mediterranean Sea, after the Sicily island. It is under the jurisdiction of **Italy**. **So, pair III is not correctly matched.**

Therefore, option (d) is the correct answer.

Q81. Consider the following statements:

Statement I:

Some rare earth elements are used in the manufacture of flat television screens and computer monitors.

Statement II:

Some rare earth elements have phosphorescent properties.

Which one of the following is correct in respect of the above statements?

- Both Statement I and Statement II are correct and Statement II explains Statement I
- Both Statement I and Statement II are correct but Statement II does **not** explain Statement I
- Statement I is correct but Statement II is not correct
- Statement I is not correct but Statement II is correct

Answer: a

Explanation:

- **Phosphorescence** is the property of certain elements of emitting light. Unlike the commonly known and applied phenomenon of fluorescence, in which the material continuously emits light falling on them, **the phosphorescent material does not re-emit the light immediately. Instead, phosphorescence releases light very slowly even after the removal of energy source.**
- **Rare Earth Metals (REM)** like europium and terbium are known for their phosphorescent properties. **So, statement 2 is correct.**
- **Phosphorescent organic light-emitting diodes (PhOLEDs) are a type of display technology that emits light using the process of phosphorescence.** They've become popular because they're much more energy-efficient than conventional technologies. In PhOLEDs, electric charges combine in an active layer to create light.

- By tweaking the materials in this layer, the color of the light can be precisely controlled—from deep blue to red—making them incredibly versatile for different applications. REMs like **europium, terbium, and yttrium** are used for creating these display screens. **So, statement I is correct.**

So, Both Statement I and Statement II are correct and Statement II explains Statement I
Therefore, option (a) is the correct answer.

Q82. Consider the following statements:

- Indian Railways have prepared a National Rail Plan (NRP) to create a 'future ready' railway system by 2028.
- 'Kavach' is an Automatic Train Protection system developed in collaboration with Germany.
- 'Kavach' system consists of RFID tags fitted on track in station section.

Which of the statements given above are **not** correct?

- I and II only
- II and III only
- I and III only
- I, II and III

Answer: a

Explanation:

- Indian Railways have prepared a **National Rail Plan (NRP) for India – 2030**. The Plan is to create a '**future ready**' Railway system by 2030. **So, statement I is not correct.**
- The objective of the Plan is to create capacity ahead of demand, which in turn would also cater to future growth in demand right up to 2050 and also increase the modal share of Railways to 45% in freight traffic and to continue to sustain it.
- Kavach** is an Automatic Train Protection system, indigenously developed by India, there is **no foreign collaboration** in the development of the same. **So, statement 2 is not correct.**
- Application of KAVACH involves placing of **Radio Frequency Identification (RFID)** throughout the track length to track train positions and detect any potential safety issues along the route. **So, statement 3 is correct.**

Therefore, option (a) is the correct answer.

Q83 Consider the following space missions:

- Axiom-4
- SpaDex
- Gaganyaan

How many of the space missions given above encourage and support micro-gravity research?

- Only one
- Only two
- All the three
- None

Answer: c

Explanation:

- Microgravity research refers to conducting experiments in the unique environment of space where the effects of weightlessness on the human body is observed. Such conditions can be both simulated on earth or can be experienced in space based missions like the International Space Station. **Axiom-4:** It is the fourth private astronaut mission to the International Space Station (ISS). It is organized by Axiom Space in collaboration with

NASA. Once docked, the private astronauts plan to spend up to 14 days aboard the ISS, conducting scientific experiments in microgravity experiments. **So, point 1 is correct.**

- **SpaDex:** It is a technology demonstrator mission for the demonstration of in-space docking using two small spacecraft launched by PSLV. **So, point 2 is correct.**
- **Gaganyaan:** It is the planned maiden manned mission of the Indian space agency, which will facilitate microgravity experiments in the Lower Earth Orbit. **So, point 3 is correct.**

So, all three of the above given space missions encourage and support micro-gravity research.

Therefore, option (c) is the correct answer.

Q84. With reference to India's defence, consider the following pairs:

Aircraft type	Description
I. Dornier-228	: Maritime patrol aircraft
II. IL-76	: Supersonic combat aircraft
III. C-17 Globe-master III	: Military transport aircraft

How many of the pairs given above are correctly matched?

- Only one
- Only two
- All the three
- None

Answer: b

Explanation:

- **Dornier 228:** It is a **twin-engine turboprop aircraft developed by Hindustan Aeronautics Limited (HAL)** in India. It is known for its lightweight design, wide range of operating speeds, and fuel efficiency. The Dornier 228 is a multi-mission aircraft used for various purposes, including **maritime patrol**, surveillance, search and rescue, and evacuation. **So, pair 1 is correctly matched.**
- **IL-76:** It is a military transport aircraft designed to **airlift troops, cargo, military equipment** and weapons, as well as conduct air dropping. **So, pair 2 is not correctly matched.**
- **C-17 Globe-master III:** It is a long range, heavy-lift strategic transport aircraft that can operate close to a potential area of operations for combat, peacekeeping or humanitarian missions worldwide. **So, pair 3 is correctly matched.**

So, only two of the above pairs are correctly matched.

Therefore, option (b) is the correct answer.

Q85. Artificial way of causing rainfall to reduce air pollution makes use of

- silver iodide and potassium iodide
- silver nitrate and potassium iodide
- silver iodide and potassium nitrate
- silver nitrate and potassium chloride

Answer: a

Explanation:

- **Clouds are suspension of water vapours in air**, these water vapors when coagulated results in downpour.

- This coagulation process can be stimulated artificially by sprinkling of ionic compounds on the clouds, most commonly used materials for the same are **silver iodide and potassium iodide**. The resultant rain is known as artificial rain.

Therefore, option (a) is the correct answer.

Q86. Consider the following statements with regard to pardoning power of the President of India:

- The exercise of this power by the President can be subjected to limited judicial review.
- The President can exercise this power without the advice of the Central Government.

Which of the statements given above is/are correct?

- I only
- II only
- Both I and II
- Neither I nor II

Answer: a

Explanation:

- **Article 72** of the Constitution of India provides for the pardoning powers to the President.
- In *Epuru Sudhakar versus the Government of Andhra Pradesh, 2006* the Supreme Court said that the powers to pardon as granted to the President under Article 72 are not immune to judicial review. **So, statement I is correct.**
- The court in this case emphasized that for effective exercise of these powers the reasons for providing or denying pardon must be mentioned. Besides, the court said that pardon obtained on the basis of manifest mistake or fraud can also be reviewed or cancelled.
- The court further elaborated that if powers under Article 72 are exercised on irrelevant discriminatory grounds or in bad faith, then in such case the judicial review becomes necessary.
- Also, in several cases, the SC had ruled that the **President has to act on advice of the council of ministers while deciding pardoning powers**, such cases include Maru Ram versus Union of India, 1980 and Dhanajay Chatterjee versus State of West Bengal. **So, statement 2 is not correct.**

Therefore, option (a) is the correct answer.

Q87. Consider the following statements:

- On the dissolution of the House of the People, the Speaker shall not vacate his/her office until immediately before the first meeting of the House of the People after the dissolution.
- According to the provisions of the Constitution of India, a Member of the House of the People on being elected as Speaker shall resign from his/her political party immediately.
- The Speaker of the House of the People may be removed from his/her office by a resolution of the House of the People passed by a majority of all the then Members of the House, provided that no resolution shall be moved unless at least fourteen days' notice has been given of the intention to move the resolution.

Which of the statements given above are correct?

- I and II only
- II and III only
- I and III only
- I, II and III

Answer: c

Explanation:

- Each House of Parliament has its own presiding officer. There is a Speaker and a Deputy Speaker for the Lok Sabha and a Chairman and a Deputy Chairman for the Rajya Sabha. A panel of chairpersons for the Lok Sabha and a panel of vice-chairpersons for the Rajya Sabha is also appointed.
- The **Speaker does not vacate her/his office with dissolution of the Lok Sabha** but vacates his office immediately **before the first meeting of the newly-elected Lok Sabha**. Therefore, the President appoints a member of the Lok Sabha as the Speaker Pro Tem. Usually, the senior most member is selected for this. **So, statement I is correct.**
- In Britain, the Speaker is strictly a non-party person because there is a convention that the Speaker has to resign from her/his party and remain politically neutral. This healthy convention is not fully established in India where the **Speaker does not resign from the membership of her/his party on her/his election to the exalted office**. **So, statement II is not correct.**
- The Speaker is provided with a security of tenure. He can be removed only by a resolution passed by the Lok Sabha by a **special majority** (ie, a **majority of all the then members of the House**) and not by an ordinary majority (ie, a majority of the members present and voting in the House). This motion of removal can be considered and discussed only when it has the support of at least 50 members. Additionally, a procedural safeguard is embedded in the process: any such resolution can be introduced only after **giving a minimum of 14 days' prior notice**, ensuring sufficient time for thorough deliberation and informed decision-making. **So, statement III is correct.**

Therefore, option (c) is the correct answer

Q88. Consider the following statements:

I. If any question arises as to whether a Member of the House of the People has become subject to disqualification under the 10th Schedule, the President's decision in accordance with the opinion of the Council of Union Ministers shall be final.

II. There is no mention of the word 'political party' in the Constitution of India.

Which of the statements given above is/are correct?

- I only
- II only
- Both I and II
- Neither I nor II

Answer: d

Explanation:

- The **Anti-Defection law** as promulgated under the **Schedule X** of the constitution of India bars the legislatures in the Parliament and the State legislative assemblies from shifting parties in the house from the party on whose nomination they were initially elected to the house. As per the act, **the Speaker of the house is the nodal agency for the execution of the same.**
- Further, this decision of Speaker is subjected to judicial review, president has no role in the same. **So, statement I is not correct.**

- The X schedule for the first provided for the constitutional status to the Political parties, wherein it mentioned that shifting of political parties in the house can lead to disqualification. **So, statement 2 is not correct.**

Therefore, option (d) is the correct answer

Q89. Consider the following statements:

Statement I:

In India, State Governments have no power for making rules for grant of concessions in respect of extraction of minor minerals even though such minerals are located in their territories.

Statement II:

In India, the Central Government has the power to notify minor minerals under the relevant law.

Which one of the following is correct in respect of the above statements?

- Both Statement I and Statement II are correct and Statement II explains Statement I
- Both Statement I and Statement II are correct but Statement II does **not** explain Statement I
- Statement I is correct but Statement II is not correct
- Statement I is not correct but Statement II is correct

Answer: d

Explanation:

In India, the minerals are classified as minor minerals and major minerals. “Minor Minerals” means building stones, gravel, ordinary clay, ordinary sand other than sand used for prescribed purposes and any other mineral that the Central Government may, by notification prescribe in the Official Gazette [According to the **Mines and Minerals (Development and Regulation) Act, 1957**].

- **The Central Government has the power to notify “minor minerals” under the MMDR Act, 1957. So, statement 2 is correct.**
- As per Section 15 of the MMDR Act, 1957, **State Governments have complete powers for making Rules for the grant of concessions in respect of extraction of minor minerals and levy and collection of royalty on minor minerals.** The power to frame policy and legislation relating to minor minerals is entirely delegated to the State Governments while policy and legislation relating to the major minerals are dealt with by the Ministry of Mines under the Union/Central Government. **So, statement I is not correct.**
- **The regulatory and administrative jurisdiction of minor minerals falls under the purview of State Governments.** These include the powers to frame rules, prescribe rates of royalty, contribution to the District Mineral Foundation, the procedure for grant of mineral concessions, regulation of their mining, control of illegal mining, etc.

So, Statement I is not correct but Statement II is correct.

Therefore, option (d) is the correct answer.

90. Which organization has enacted the Nature Restoration Law (NRL) to tackle climate change and biodiversity loss?

- The European Union
- The World Bank
- The Organization for Economic Cooperation and Development

d) The Food and Agriculture Organization

Answer: a

Explanation:

Nature Restoration Law (NRL):

- It is a **European Union (EU)** law to tackle the triple crises of climate change, biodiversity loss, and environmental degradation.
- It is the **first continent-wide, comprehensive law of its kind**. It is a key element of the EU Biodiversity Strategy, which sets binding targets to restore degraded ecosystems, in particular those with the most potential to capture and store carbon and to prevent and reduce the impact of natural disasters.
- Under the NRL, **Member States will have to restore at least 20% of the EU's land** and sea by 2030, including terrestrial, coastal, and freshwater, forest, agricultural, and urban areas.
- By 2050, the measures should extend to all ecosystems "requiring restoration". The EU NRL also targets an obligation to improve urban green spaces, contribute to free-flowing rivers by removing artificial barriers, increase pollinator populations, and contribute to the target of 3 billion additional trees throughout the EU.
- Member States must adopt "restoration plans" detailing how they intend to achieve these targets and ensure that the restored areas do not significantly deteriorate.
- While drafting these Nature Restoration Plans, **Member States also need to consider socio-economic impacts** and benefits and estimate the financial needs for its implementation.

Therefore, option (a) is the correct answer.



बैच 2

**स्टेपअप मेंटरशिप
प्रोग्राम 2026**

शुल्क: ₹ 40000 15% छूट के साथ

शुरू: 14 मई 2025

मोड: ऑफ़लाइन और ऑनलाइन

Q91. Consider the following statements:

I. Panchayats at the intermediate level exist in all States.

II. To be eligible to be a Member of a Panchayat at the intermediate level, a person should attain the age of thirty years.

III. The Chief Minister of a State constitutes a commission to review the financial position of Panchayats at the intermediate levels and to make recommendations regarding the distribution of net proceeds of taxes and duties, leviable by the State, between the State and Panchayats at the intermediate level.

Which of the statements given above are **not** correct?

- a) I and II only
- b) II and III only
- c) I and III only
- d) I, II and III

Answer: d

Explanation:

- **Article 243B:** Panchayats at the intermediate level may not be constituted in a State having a population not exceeding **twenty lakhs**. **So, statement I is not correct.**
- **Article 243I:** The **Governor of a State** shall, as soon as may be within one year from the commencement of the Constitution (Seventy-third Amendment) Act, 1992, and thereafter at the expiration of every fifth year, constitute a **Finance Commission** to review the financial position of the Panchayats and to make recommendations. **So, statement 3 is not correct.**
- **Article 243F:** No person shall be disqualified on the ground that he is less than twenty-five years of age, if he has attained the age of **twenty-one years**. **So, statement 2 is not correct.**

Therefore, option (d) is the answer.

Q92. Consider the following statements in respect of BIMSTEC:

I. It is a regional organization consisting of seven member States till January 2025.

II. It came into existence with the signing of the Dhaka Declaration, 1999.

III. Bangladesh, India, Sri Lanka, Thailand and Nepal are founding member States of BIMSTEC.

IV. In BIMSTEC, the sub sector of 'tourism' is being led by India.

Which of the statements given above is/are correct?

- a) I and II
- b) I and III
- c) I and IV
- d) I only

Answer: d

Explanation:

- The Bay of Bengal Initiative for Multi-Sectoral Technical and Economic Cooperation (BIMSTEC) is a grouping of **seven Member States lying in the littoral and adjacent areas of the Bay of Bengal**. It is a unique link connecting South Asia with South-East Asia - five Members from South Asia (Bangladesh,

Bhutan, India, Nepal and Sri Lanka) and two from South-East Asia (Myanmar and Thailand). **So, statement 1 is correct.**

- **BIMSTEC** was formed in **1997**. It was established with the signing of the **Bangkok Declaration**. **So, statement 2 is not correct.**
- It was **originally formed with four Member States** with the acronym 'BIST-EC' (**Bangladesh, India, Sri Lanka and Thailand Economic Cooperation**). Following inclusion of Myanmar in December 1997, it was renamed 'BIMST-EC' (Bangladesh, India, Myanmar, Sri Lanka and Thailand Economic Cooperation). With the inclusion of Nepal and Bhutan in 2004, the name of the grouping was changed to BIMSTEC. **So, statement 3 is not correct.**
- **Nepal**, as the Lead Country of 'Tourism' has circulated a draft of **"Plan of Action on Tourism"** updating on the earlier plans, which has to be discussed and finalized during the upcoming meetings of tourism Sub-sector. **So, statement 4 is not correct.**

Therefore, option (d) is the correct answer.

Q93. Who amongst the following are members of the Jury to select the recipient of 'Gandhi Peace Prize'?

- The President of India
- The Prime Minister of India
- The Chief Justice of India
- The Leader of Opposition in the Lok Sabha

Select the correct answer using the code given below.

- II and IV only
- I, II and III
- II, III and IV
- I and III only

Answer: c

Explanation:

Gandhi Peace Prize:

- Instituted in 1995, the Gandhi Peace Prize is awarded for **social, economic and political transformation through non-violence**, to any deserving person/s or institution/s.
- The award comprises a citation and an amount of Rs 1 crore and is open to all persons regardless of nationality, race, language, caste, creed or gender and any association, institution or organisation.
- The awardee is selected by a five-member jury comprising of the:
 - Prime Minister of India. **So, point 2 is correct.**
 - Chief Justice of India. **So, point 3 is correct.**
 - Leader of Opposition. **So, point 4 is correct.**
 - Two eminent personalities.
- Nominees must be proposed either by former jury members, former awardees, members of Parliament, Nobel laureates for the last five years, Secretary-General of the United Nations or heads of other international peace organisations, Vice-Chancellors of universities, Chief Ministers, or Governors. Work achieved within the ten years preceding the nomination is considered for the award. The decision of the jury is final and cannot be challenged or appealed.

- The **President of India is not a member** of the jury to select the recipient of 'Gandhi Peace Prize.
So, point I is not correct.

Therefore, option (c) is the correct answer.

Q94. GPS-Aided Geo Augmented Navigation (GAGAN) uses a system of ground stations to provide necessary augmentation. Which of the following statements is/are correct in respect of GAGAN?

- It is designed to provide additional accuracy and integrity.
- It will allow more uniform and high quality air traffic management.
- It will provide benefits only in aviation but not in other modes of transportation.

Select the correct answer using the code given below.

- I, II and III
- II and III only
- I only
- I and II only

Answer: d

Explanation:

- **GPS Aided GEO Augmented Navigation (GAGAN)** uses a system of ground stations to provide necessary augmentations to the GPS standard positioning service (SPS) navigation signal.
- The GAGAN is designed to provide the **additional accuracy, availability, and integrity necessary to enable users to rely on GPS for all phases of flight**, from en route through approach for all qualified airports within the GAGAN service volume. **So, statement I is correct.**
- GAGAN will also provide the capability for **increased accuracy in position reporting, allowing for more uniform and high-quality Air Traffic Management (ATM)**. **So, statement 2 is correct.**
- In addition, GAGAN will provide **benefits beyond aviation to all modes of transportation, including maritime, highways, and railroads**. **So, statement 3 is not correct.**
- A network of precisely surveyed ground reference stations (INdian Reference Stations INRES) is strategically positioned across the country to collect GPS satellite data. Using this information, the master control centre (INdian Master Control Centre INMCC) generates messages to correct any signal errors. These correction messages are then uplinked through (INdian Land Uplink Station INLUS) and broadcast through communication satellites (Geostationary) to receivers onboard aircraft using the same frequency as GPS.

Therefore, option (d) is the correct answer.

Q95. Consider the following statements regarding AI Action Summit held in Grand Palais, Paris in February 2025:

- Co-chaired with India, the event builds on the advances made at the Bletchley Park Summit held in 2023 and the Seoul Summit held in 2024.
 - Along with other countries, the US and UK also signed the declaration on inclusive and sustainable AI.
- Which of the statements given above is/are correct?

- I only
- II only
- Both I and II

d) Neither I nor II

Answer: a

Explanation:

- On 10 and 11 February 2025, France and India co-chaired the Artificial Intelligence Action Summit, gathering Heads of State and Government, leaders of international organizations, small and large enterprises, representatives of academia, non-governmental organizations, artists and members of civil society, in order to **build on the important milestones reached during the Bletchley Park (November 2023) and Seoul (May 2024) summits. So, statement 1 is correct.**
- Both the countries underlined their commitment to take concrete actions to ensure that the global AI sector can drive beneficial social, economic and environmental outcomes in the public interest.
- The **US and the UK have refused to sign a declaration** on “inclusive and sustainable” artificial intelligence in a blow to hopes for a concerted approach to developing and regulating the technology. **So, statement 2 is not correct.**

Therefore, option (a) is the correct answer.

Q96. Consider the following pairs:

I	International Year of the Woman Farmer	:	2026
II.	International Year of Sustainable and Resilient Tourism	:	2027
III.	International Year of Peace and Trust	:	2025
IV.	International Year of Asteroid Awareness and Planetary Defence	:	2029

How many of the pairs given above are correctly matched?

- Only one
- Only two
- Only three
- All the four

Answer: d

Explanation:

- The Food and Agriculture Organization of the United Nations (FAO) recently welcomed the resolution adopted by the United Nations (UN) General Assembly to declare **2026 as the International Year of the Woman Farmer.** The resolution, which was proposed by the United States of America and adopted by consensus, invites FAO, in collaboration with the other UN Rome-based Agencies, to facilitate the implementation and observance of the International Year of the Woman Farmer. **So, pair 1 is correctly matched.**
- The UN General Assembly decided to proclaim **2027 the International Year of Sustainable and Resilient Tourism.** It encouraged all states, the United Nations system and all other actors to take advantage of the International Year to promote actions at all levels, including through international cooperation, and to support sustainable and resilient tourism as a means of promoting and accelerating sustainable development. **So, pair 2 is correctly matched.**

- In 2024, the UN General Assembly declared **2029 the International Year of Asteroid Awareness and Planetary Defence** to take advantage of the close approach of 99942 Apophis and raise global awareness about asteroids. **So, pair 4 is correctly matched.**
 - This initiative aims to highlight the collaborative efforts of the Committee on the Peaceful Uses of Outer Space in mitigating potential hazards from near-Earth objects, while also providing an opportunity for a global educational campaign.
 - On 13 April 2029, the asteroid 99942 Apophis will pass safely at a distance of about 32,000 kilometers above Earth's surface, within the geostationary orbit, posing no threat to the planet. This extremely close approach will make the asteroid visible to billions of people with the naked eye in the clear night sky.
- Violent conflicts are at their highest levels since the United Nations was founded 80 years ago. To strengthen a global culture of peace and improve the lives of people everywhere, **UN Member States have declared 2025 the International Year of Peace and Trust.** The General Assembly Resolution calls on the international community to redouble efforts to prevent and resolve conflicts through dialogue and diplomacy. **So, pair 3 is correctly matched.**

Therefore, option (d) is the correct answer.

Q97. Consider the following statements with regard to BRICS :

I. 16th BRICS Summit was held under the Chairship of Russia in Kazan.

II. Indonesia has become a full member of BRICS.

III. The theme of the 16th BRICS Summit was Strengthening Multiculturalism for Just Global Development and Security.

Which of the statements given above is/are correct?

- a) I and II
- b) II and III
- c) I and III
- d) I only

Answer: a

Explanation:

- The **16th BRICS Summit** concluded in October 2024, with the adoption of the Kazan Declaration, a comprehensive document **outlining key areas of cooperation and the bloc's unified stance on global issues.**
- The Summit held in **Kazan, Russia**, saw productive discussions on strengthening multilateralism, countering terrorism, promoting economic growth and addressing the concerns of the Global South. **So, statement I is correct.**
- Under the theme "**Strengthening Multilateralism for Just Global Development and Security**", the Kazan Declaration emphasised the importance of BRICS solidarity and strategic partnerships. The leaders pledged their commitment to fostering peace, ensuring a fairer international order, and promoting sustainable development. **So, statement 3 is not correct.**
- Brazil recently announced the official entry of **Indonesia as a full member of BRICS, marking the first expansion of the group under the Brazilian presidency**, which began on January 1. Indonesia's membership was approved at the Johannesburg Summit in August 2023 and formalized after the formation of the country's new government, which waited for the conclusion of the presidential elections in 2024 to confirm its interest. **So, statement 2 is correct.**

Therefore, option (a) is the correct answer.

Q98. Consider the following statements about Lokpal:

- I. The power of Lokpal applies to public servants of India, but not to the Indian public servants posted outside India.
- II. The Chairperson or a Member shall not be a Member of the Parliament or a Member of the Legislature of any State or Union Territory, and only the Chief Justice of India, whether incumbent or retired, has to be its Chairperson.
- III. The Chairperson or a Member shall not be a person of less than forty-five years of age on the date of assuming office as the Chairperson or Member, as the case may be.
- IV. Lokpal cannot inquire into the allegations of corruption against a sitting Prime Minister of India.
- Which of the statements given above is/are correct?

- a) III only
b) II and III
c) I and IV
d) None of the above statements is correct

Answer: a

Explanation:

- **Lokpal and Lokayuktas Act, 2013** provides for the institution of the Lokpal at the Centre and the Lokayukta at the level of the State. The law extends to the whole of India and applies to **public servants both in and outside India**.
- Following is the jurisdiction of Lokpal:
 - Any person who is or has been a **Prime Minister**. **So, statement 4 is not correct.**
 - All categories of public servants, including Group A, Group B, Group C, and Group D officers and employees of the Government.
 - Any person who is or has been a **Minister of the Union**.
 - The definition of a public servant **does not include** those who come under the **Army Act, Navy Act, Air Force Act, or Coast Guard Act**.
- The Lokpal shall consist of— (a) a Chairperson, who is or has been a Chief Justice of India or is or **has been a Judge of the Supreme Court** or an eminent person. **So, statement 2 is not correct.**



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- The Chairperson or a Member shall **not be a person of less than forty-five years of age**, on the date of assuming office as the Chairperson or Member, as the case may be. **So, statement 3 is correct.**
- The **Lokpal and Lokayuktas Act, 2013** applies to public servants in and outside India. **So, statement 1 is not correct.**

Therefore, option (a) is the correct answer.

Q99. Consider the following statements in respect of the first Kho Kho World Cup:

I. The event was held in Delhi, India.

II. Indian men beat Nepal with a score of 78-40 in the final to become the World Champion in men category.

III. Indian women beat Nepal with a score of 54-36 in the final to become the World Champion in women category.

Which of the statements given above is/are correct?

- I only
- II and III only
- I and III only
- I, II and III

Answer: a

Explanation:

- In January 2025, the inaugural **Kho Kho World Cup in New Delhi** concluded on a historic note with India securing the titles in both men's and women's competitions. **So, statement I is correct.**
- While the Indian men's kho kho team beat Nepal 54-36 in the title clash, the women's team outplayed the Nepalese eves by a 78-40 scoreline. **So, statements 2 and 3 are not correct.**

Therefore, option (a) is the correct answer.

Q100. Consider the following statements:

I. In the finals of the 45th Chess Olympiad held in 2024, Gukesh Dommaraju became the world's youngest winner after defeating the Russian player Nepomniachtchi.

II. Abhimanyu Mishra, an American chess player, holds the record of becoming world's youngest ever Grandmaster.

Which of the statements given above is/are correct?

- I only
- II only
- Both I and II
- Neither I nor II

Answer: b

Explanation:

- **Gukesh Dommaraju** has made history by becoming the youngest chess world champion after defeating defending champion **Ding Liren of China** in the final match of their series held in Singapore. **So, statement I is not correct.**
- **Abhimanyu Mishra from America** became the youngest international master in 2019 at the age of 10 and began earning his GM norms in 2021. **So, statement 2 is correct.**

Therefore, option (b) is the correct answer.